

Time-series Temporal Dependence & ARIMA

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POLITECNICO
MILANO 1863

Reflecting ...



Fact

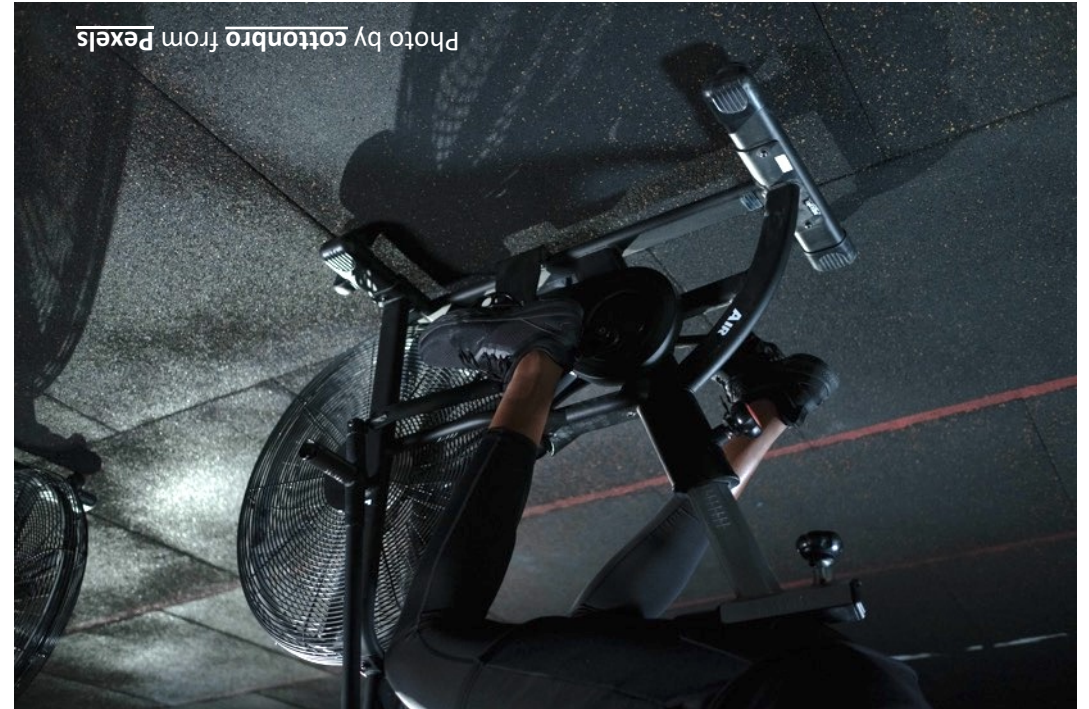
If a time series is stationary,... ... it is predictable



Question

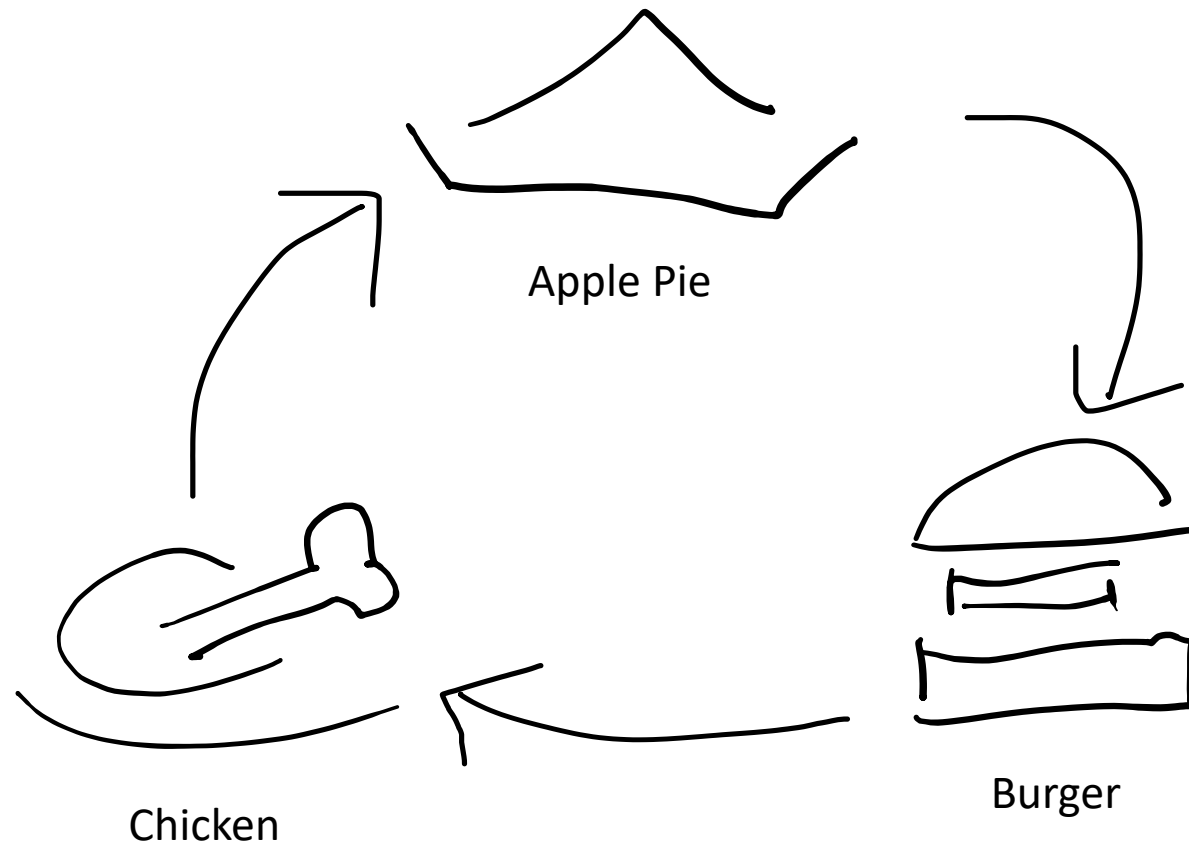


If a time series is predictable, ...

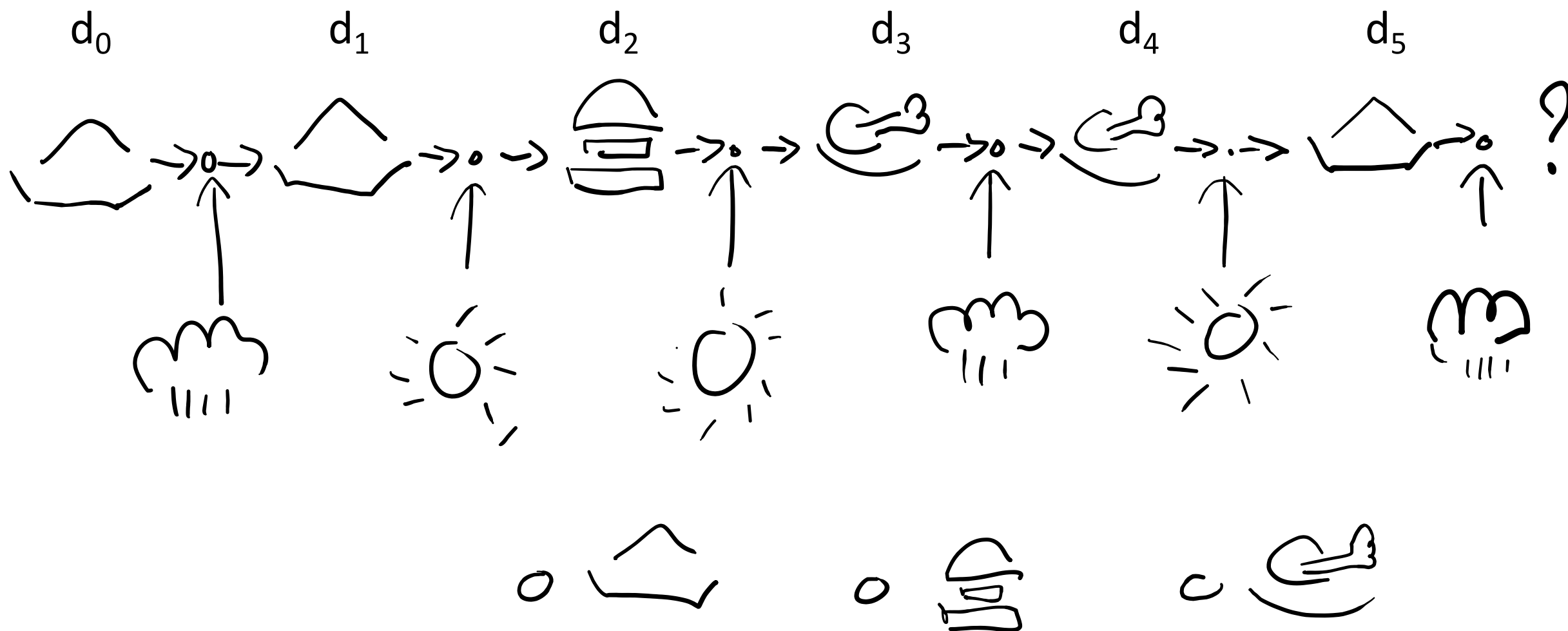


..., is it then stationary?

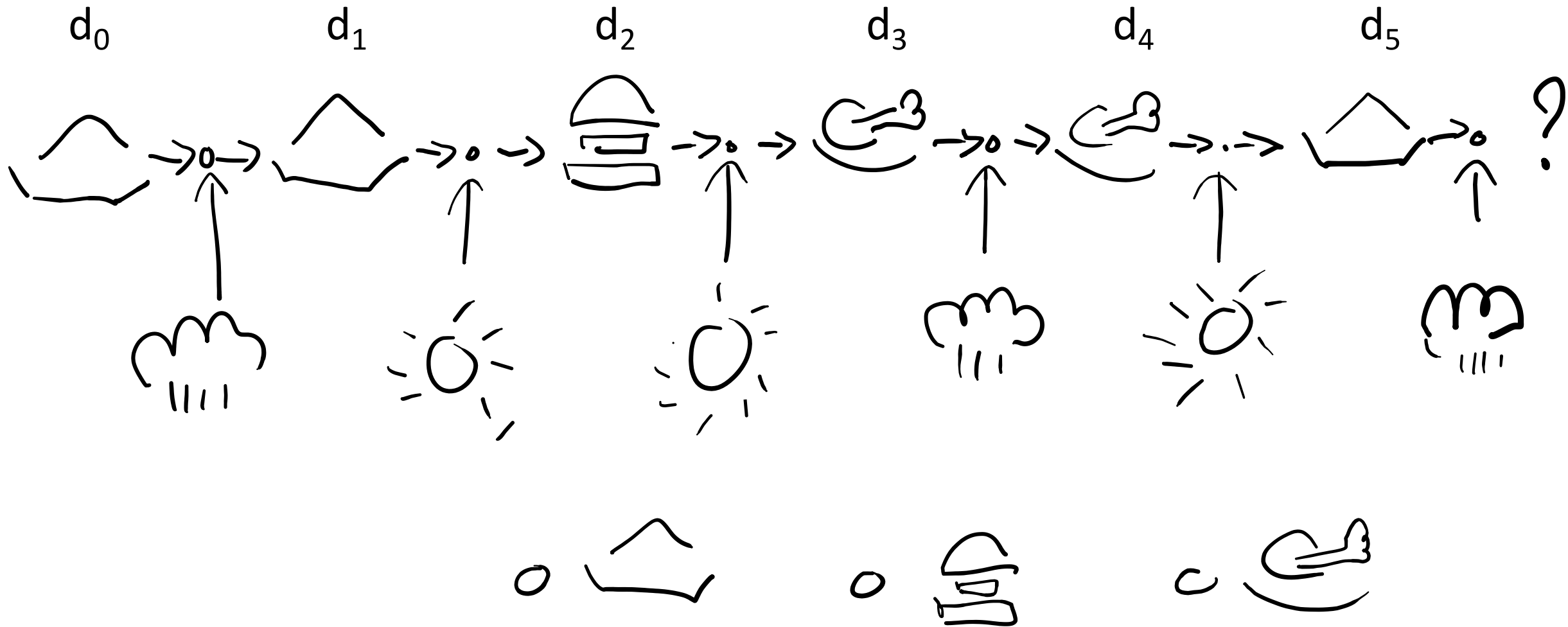
E.g., methodic but meteosensitive chef



E.g., methodic but meteosensitive chef



E.g., methodic but meteosensitive chef



Building the intuition

- Why is the time series (a.k.a., sequence) predictable in this case?

Naïve vs DES vs TES

Naïve beats DES

TES performs barely better than Naïve

Why?



2nd foundational concept

Temporal Dependence

How do we find temporal dependence?

There are few concept to recall/understand:

- Correlation
- Autocorrelation
- Autocorrelation Function
- Partial Autocorrelation Function

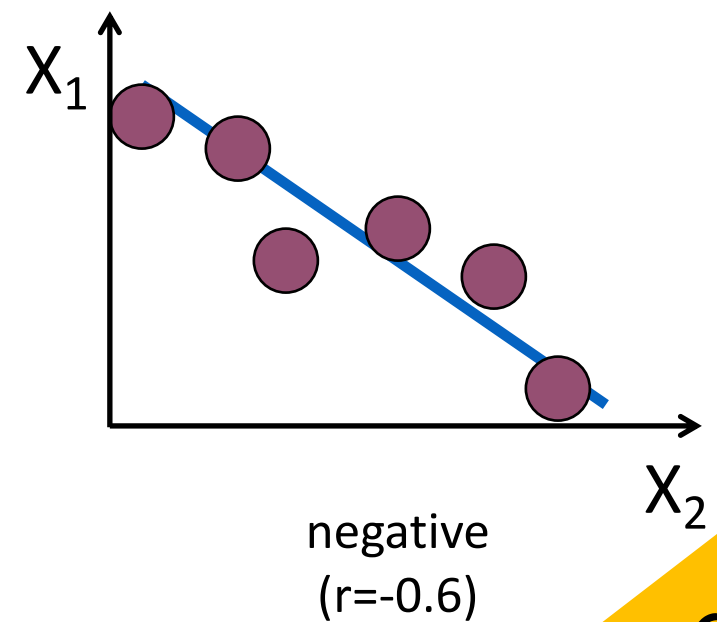
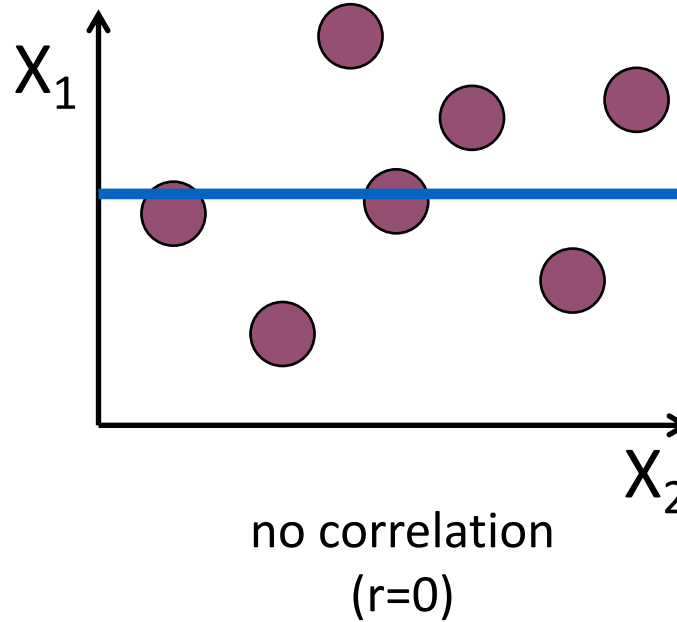
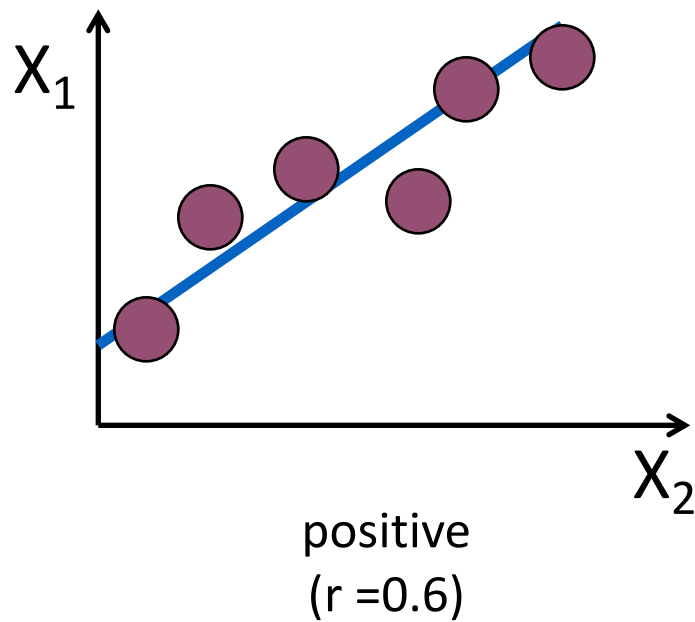


Let's build the
intuition of
**temporal
dependency**



Correlation

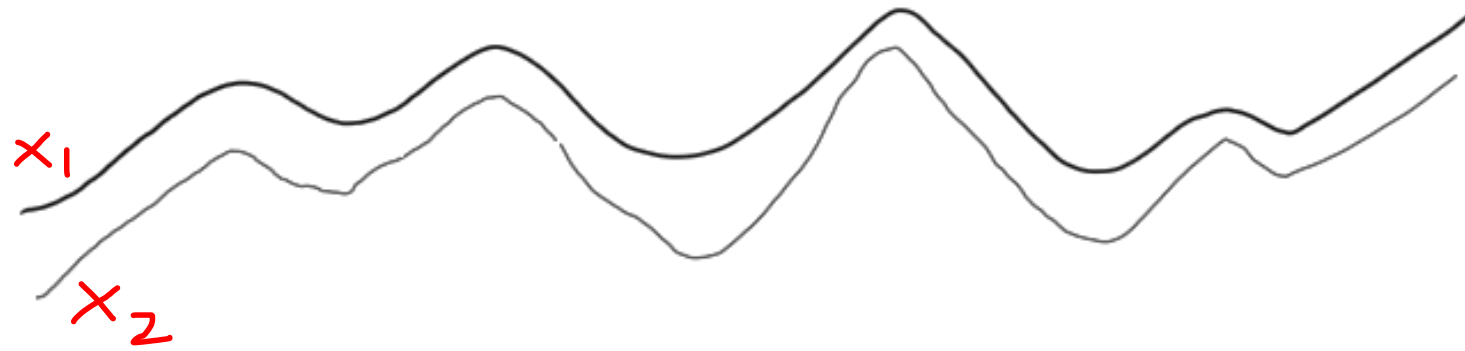
- **Correlation** describes the **degree to which two variables X_1 and X_2 move in coordination** with one another



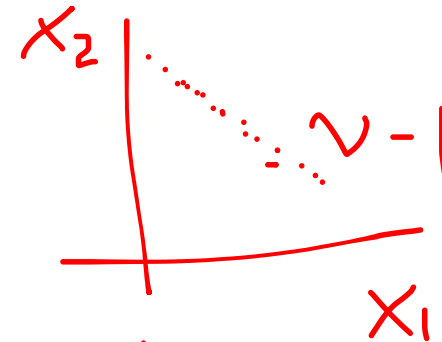
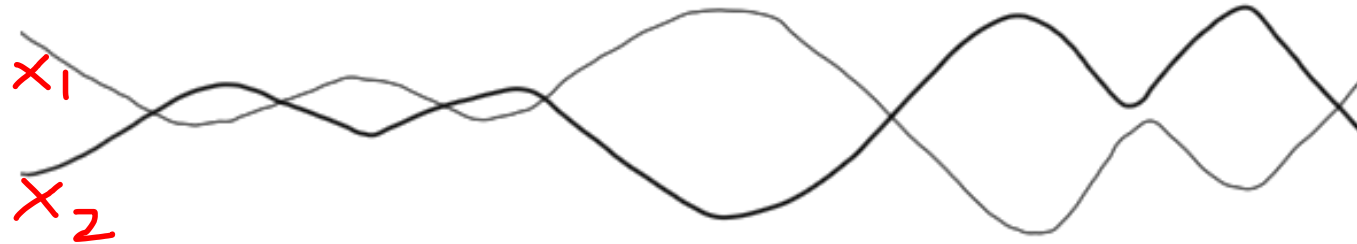
Recall

Comparing two lines over time

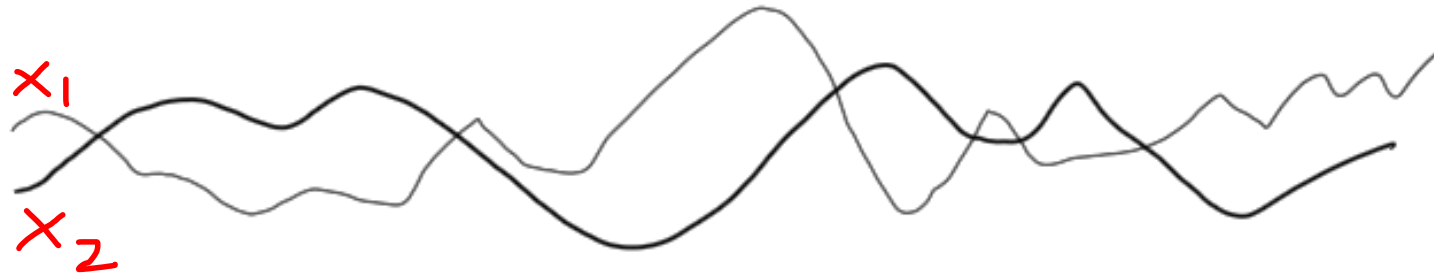
Positive correlation



Negative correlation

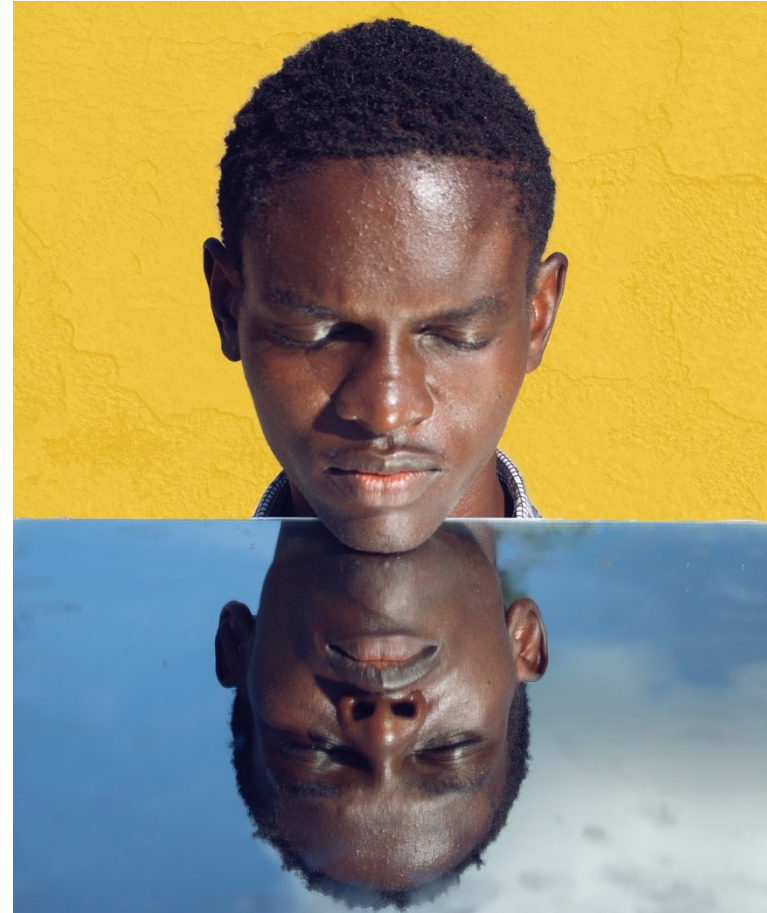


No correlation



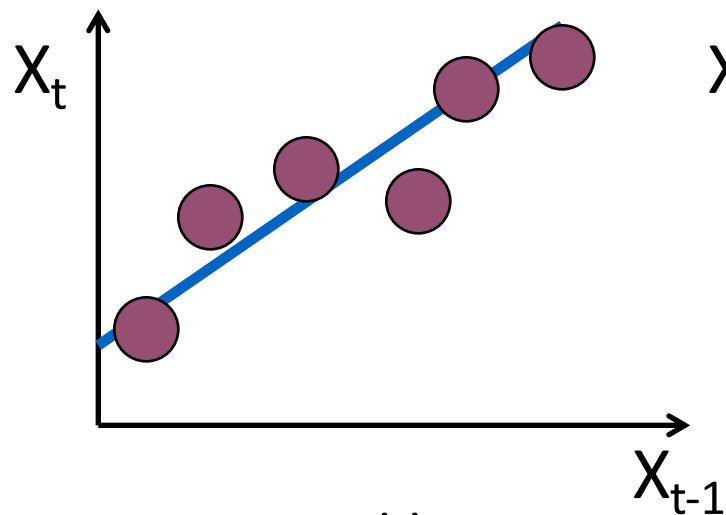
Recall

Intuitively Correlation vs. autocorrelation

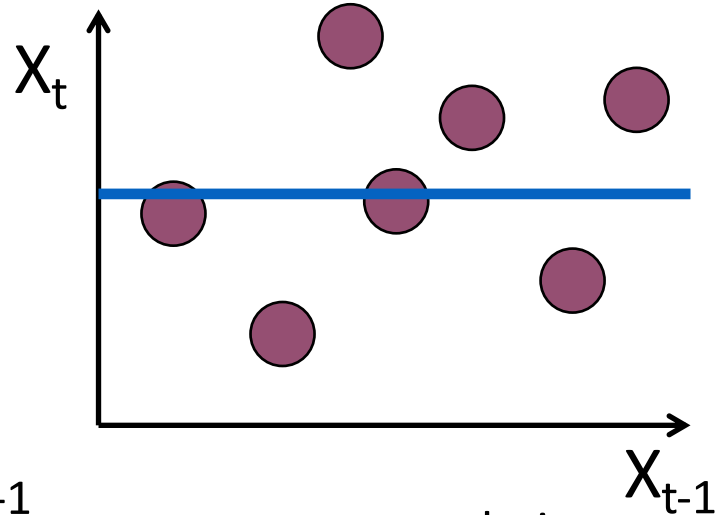


Autocorrelation at lag 1

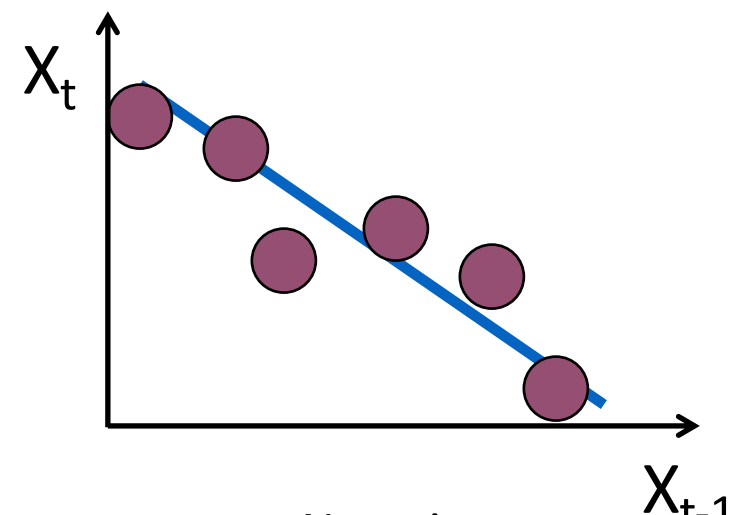
- **Autocorrelation at lag 1** describes the degree of **similarity between** a given **time series** and a **version of itself lagged once**.



positive
autocorrelation
with lag=1



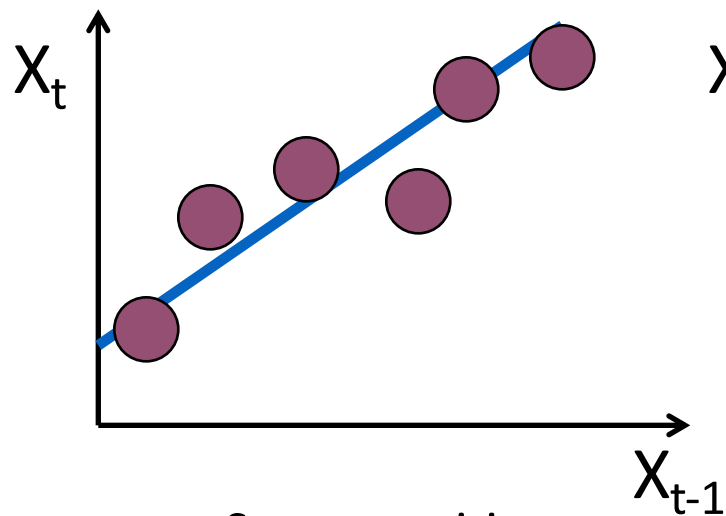
no autocorrelation
with lag=1



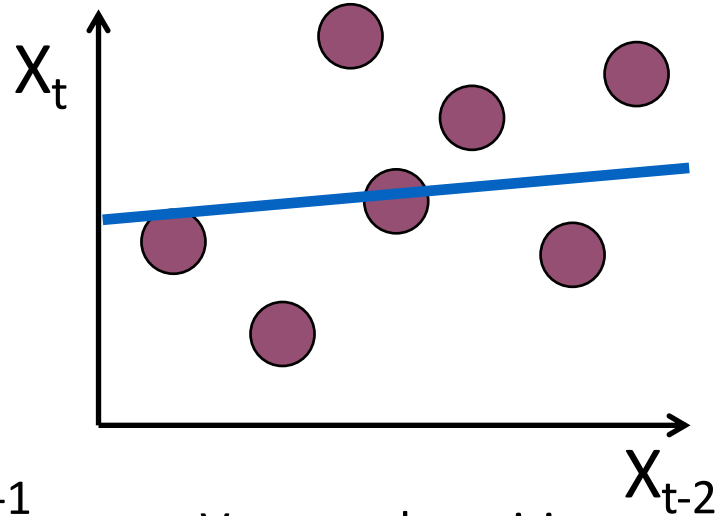
Negative
autocorrelation
with lag=1

Autocorrelation at lag k

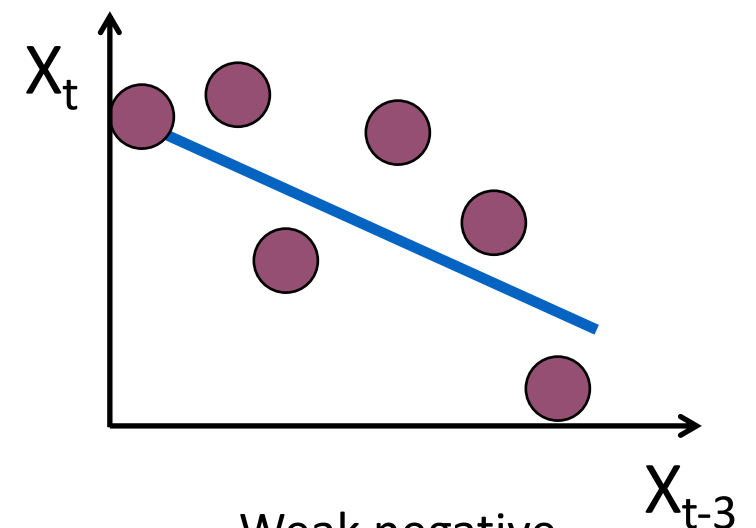
- **Autocorrelation** describes the degree of **similarity between** a given **time series and a lagged version** of itself **over successive time intervals**.



Strong positive
autocorrelation
with lag=1

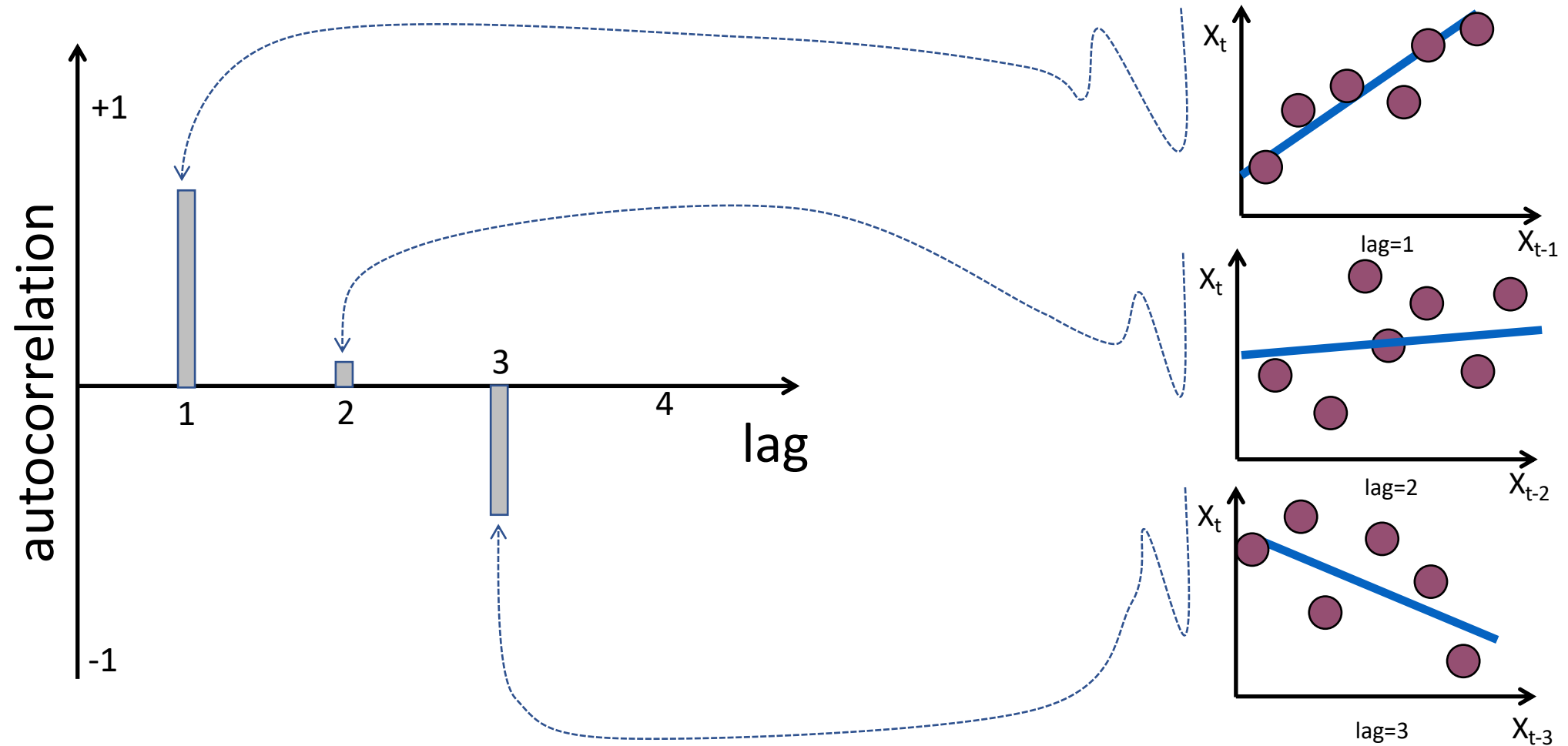


Very weak positive
with lag=2



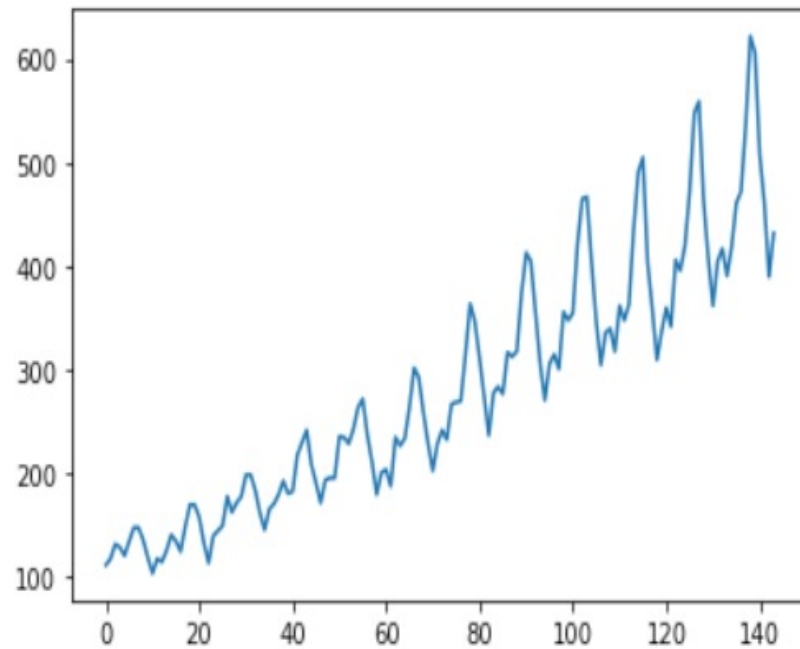
Weak negative
autocorrelation
with lag=3

Autocorrelation Function (ACF)

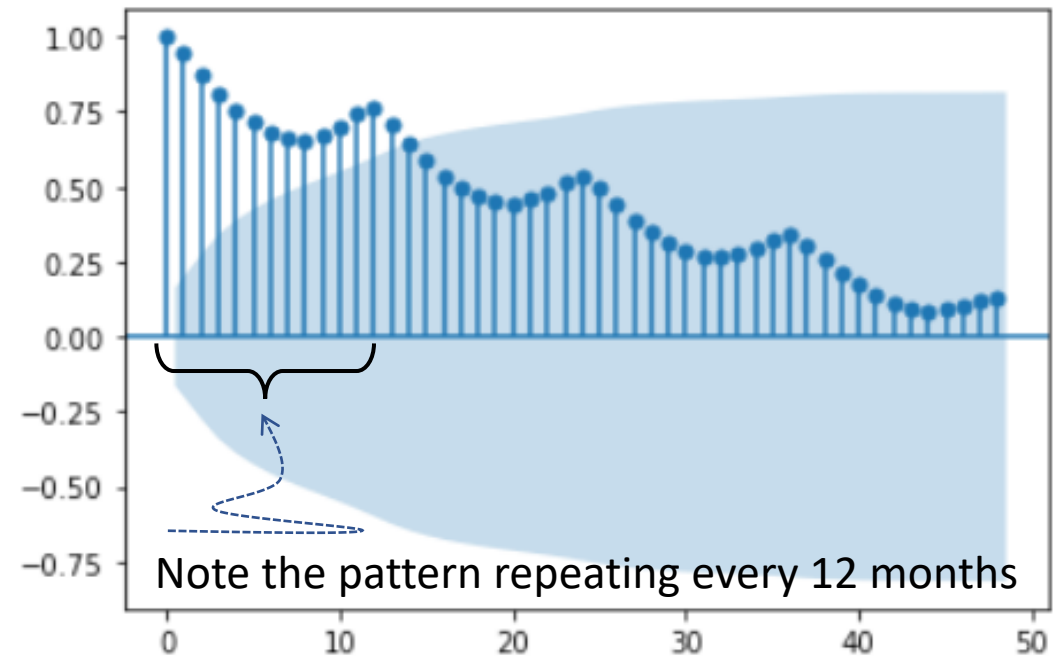


E.g., autocorrelation function for air passengers

Time series



ACF



NOTE: The **blue area** depicts the 95% confidence interval and is an indicator of the significance threshold. Anything within the blue area is statistically close to zero and anything outside the blue area is statistically non-zero.

I Got it!

A time series is not stationary if it has a seasonal component because **seasonality** is a special type of **temporal dependence**!



Isn't that

*The presence of autocorrelation... ... makes a time series **more** predictable?*

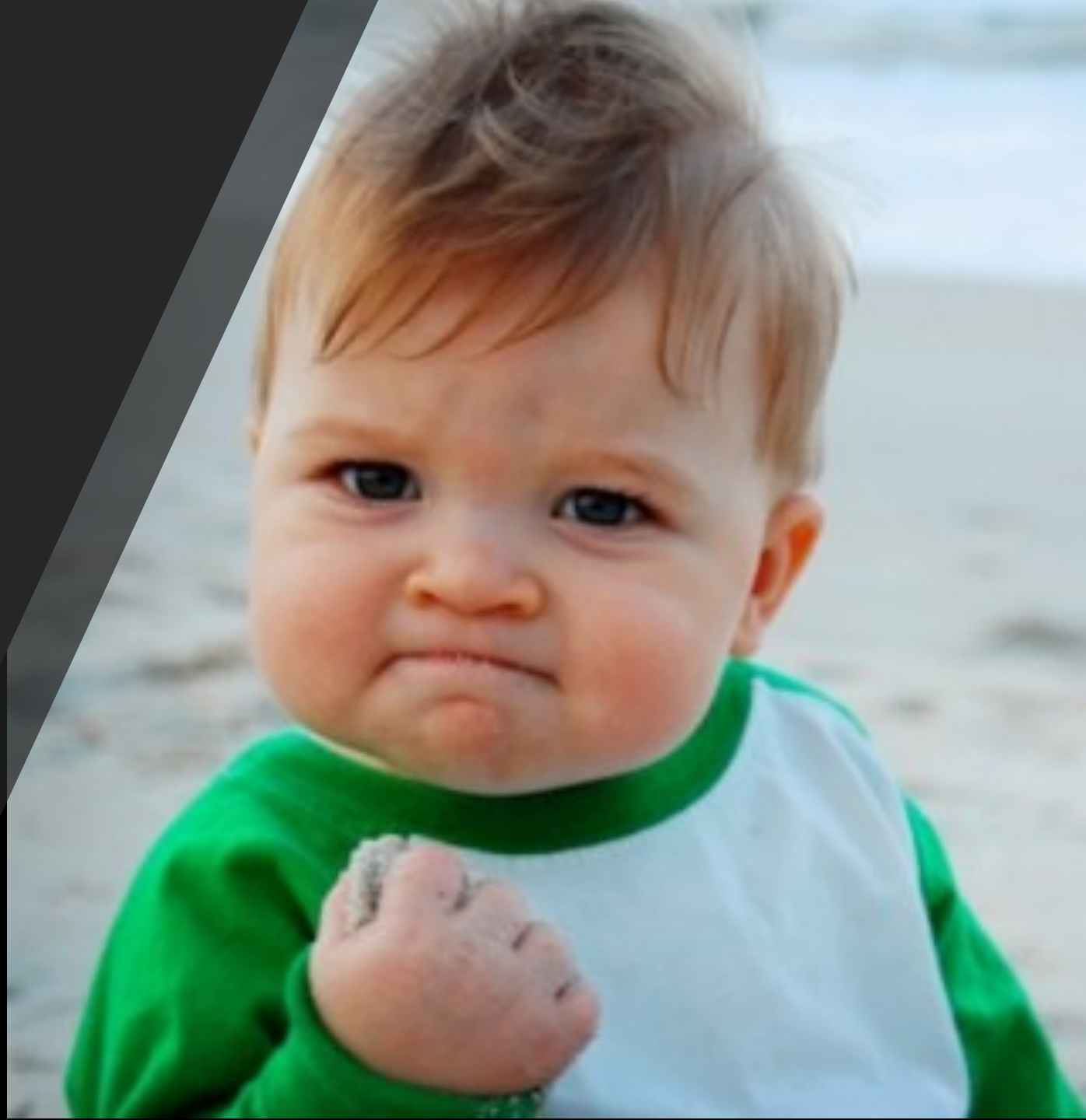


BALLA, Dog and person in Movement



Photo by [Bogdan Dirică](#) from [Pexels](#)

Oh yeah!



Formally...



Photo by [George Becker](#) from [Pexels](#)

ARMA models

- An **ARMA** model of order **p**, **q**, denoted as **ARMA(p,q)**, is the **sum of**
 - an **Auto Regressive (AR) model** of p previous of the values **and**
 - a **Moving Average (MA) model** of q previous estimation errors

$$\begin{aligned} y_t = & c + \alpha_1 y_{t-1} + \alpha_2 y_{t-2} + \dots + \alpha_p y_{t-p} \quad \left. \vphantom{c + \alpha_1 y_{t-1} + \alpha_2 y_{t-2} + \dots + \alpha_p y_{t-p}} \right\} \text{AR}(p) \\ & + \\ & \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q} \quad \left. \vphantom{\epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q}} \right\} \text{MA}(q) \end{aligned}$$

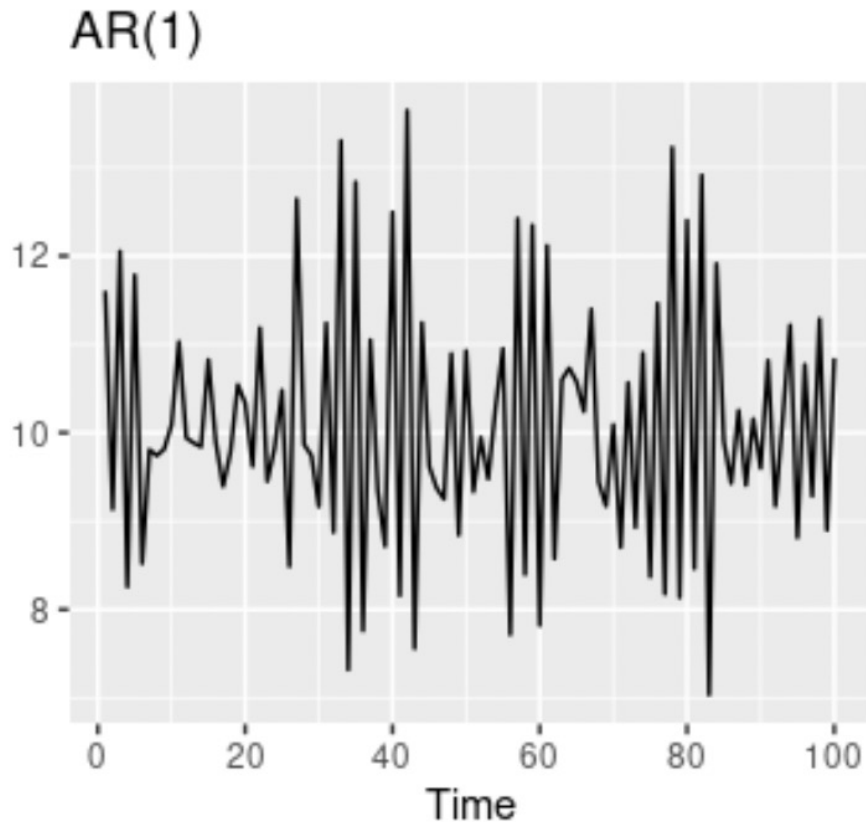
Autoregressive models

- An **autoregressive** model **of order p** , denoted as **AR(p)**, models a time series so that the current value of a time series is a linear combination of the **past p values of the time series** and white noise ϵ_t

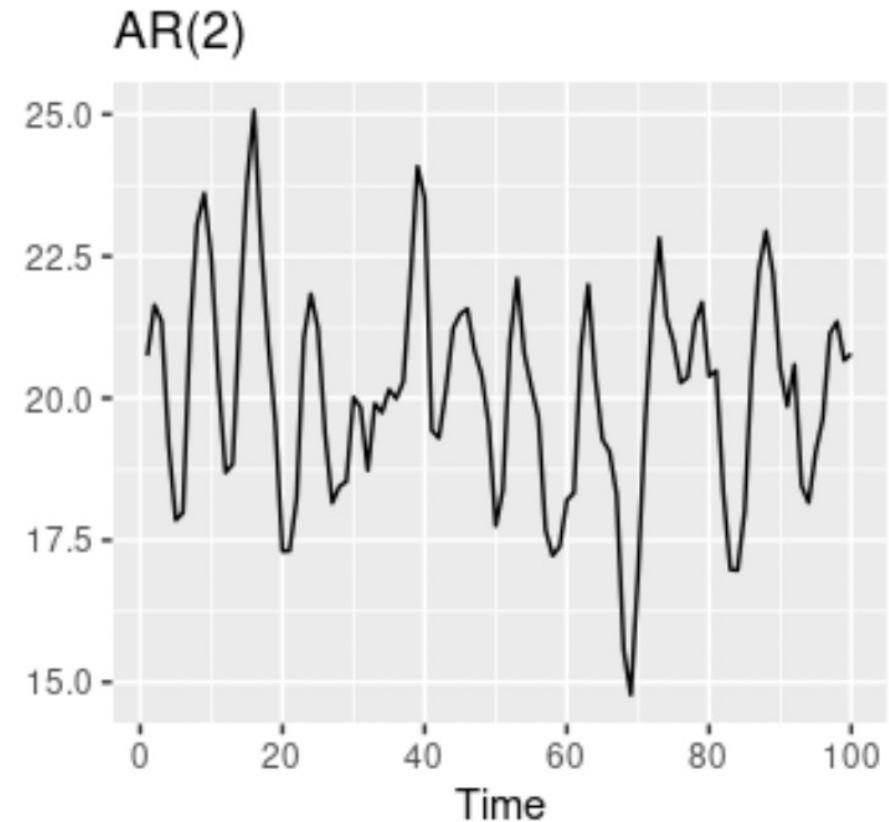
$$y_t = c + \alpha_1 y_{t-1} + \alpha_2 y_{t-2} + \dots + \alpha_p y_{t-p} + \epsilon_t = c + \sum_{n=1}^p \alpha_n y_{t-n} + \epsilon_t$$

- Notes:
 - It is similar to Exponential Smoothing, but it uses only p values and the weights α_i do not decay exponentially
 - It can improve forecast with a forecast horizon smaller than p (i.e., short-term)

Example of autoregressive models



$$y_t = 18 - 0.8y_{t-1} + \varepsilon_t$$



$$y_t = 8 + 1.3y_{t-1} - 0.7y_{t-2} + \varepsilon_t$$

Some special AR(1) models

- When $c = \alpha_1 = 0$, AR(1) is equivalent to white noise
- When $c = 0$ and $\alpha_1 = 1$, AR(1) is equivalent to random walk
- When $c \neq 0$ and $\alpha_1 = 1$, AR(1) is equivalent to random walk with drift
- When $c = 0$ and $\alpha_1 < 1$, AR(1) tends to oscillate between positive and negative values

Moving Average models

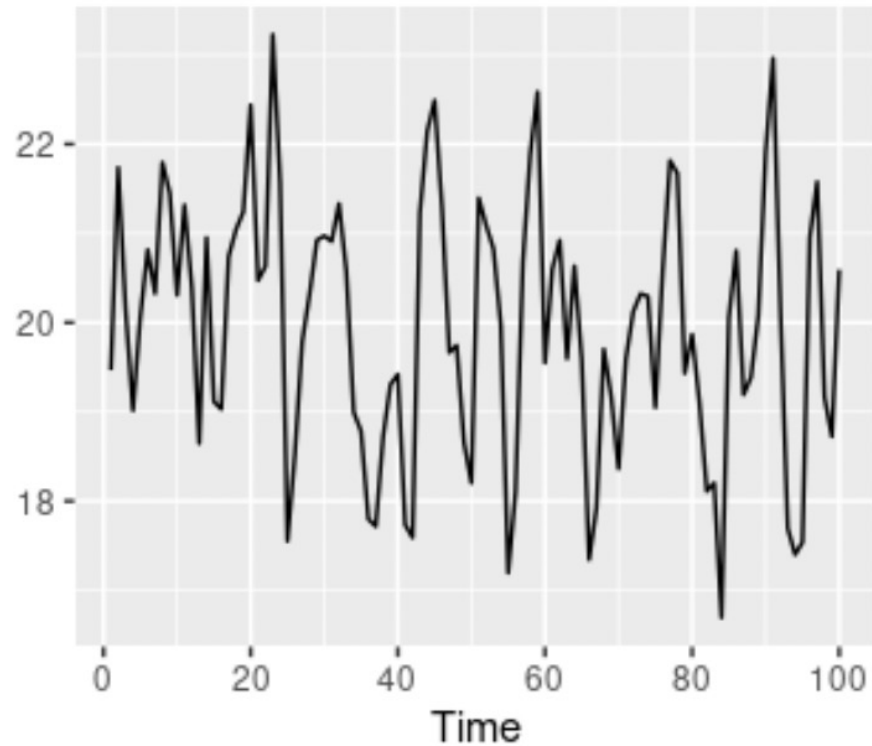
- A **moving average** model **of order q**, denoted as **MA(q)**, models the residual of a time series using **regression of past q estimation errors**

$$y_t = c + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_p \epsilon_{t-q} = c + \sum_{n=1}^q \theta_n \epsilon_{t-n} + \epsilon_t$$

- NOTE: Not be confused MA models with the moving average of the time series value! The MA models works on estimation errors

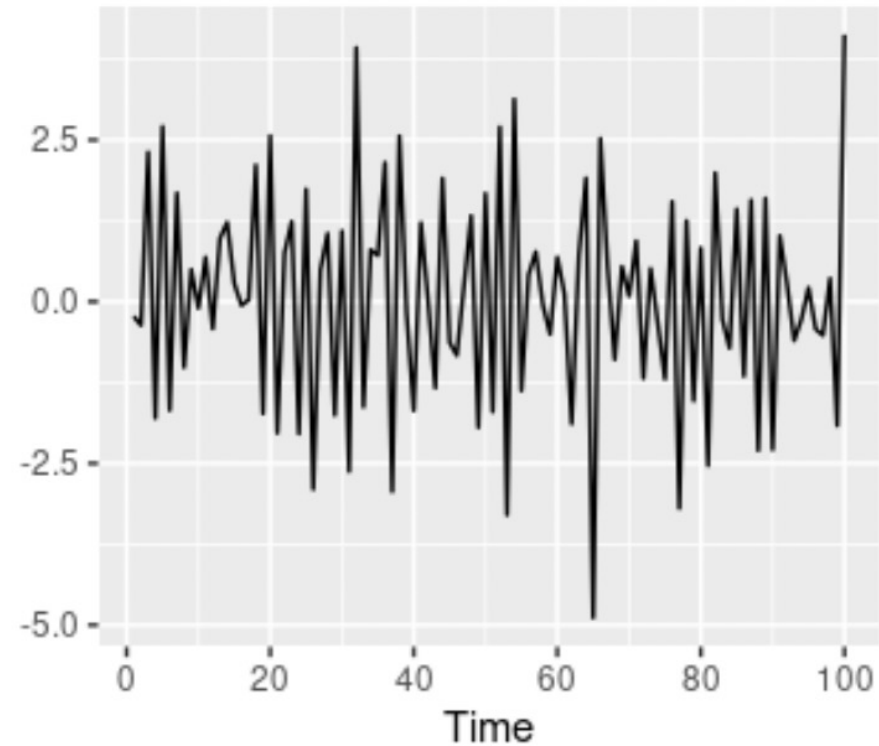
Example of MA models

MA(1)



$$y_t = 20 + \varepsilon_t + 0.8\varepsilon_{t-1}$$

MA(2)



$$y_t = \varepsilon_t - \varepsilon_{t-1} + 0.8\varepsilon_{t-2}$$

From AR(p) to MA(∞)

It is possible to write any **stationary** AR(p) model as an MA(∞) model. For example, using repeated substitution, we can demonstrate this for an AR(1) model:

$$\begin{aligned}y_t &= \alpha_1 y_{t-1} + \epsilon_t \\&= \alpha_1 (\alpha_1 y_{t-2} + \epsilon_{t-1}) + \epsilon_t \\&= \alpha_1^2 y_{t-2} + \alpha_1 \epsilon_{t-1} + \epsilon_t \\&= \alpha_1^3 y_{t-3} + \alpha_1^2 \epsilon_{t-2} + \alpha_1 \epsilon_{t-1} + \epsilon_t \\&\dots\end{aligned}$$

Provided $-1 < \alpha_1 < 1$, the value of α_1^k will get smaller as k gets larger. So eventually we obtain the above equation equal to a MA(∞) process.

ARIMA: ARMA + differencing

- An **ARIMA** model of order **(p, d, q)** is the **sum of**
 - an **Auto Regressive (AR)** model of p previous ***lagged*** values and
 - a **Moving Average (MA)** model of q previous ***lagged*** estimation errors

$$\begin{aligned} y'_t = c + \alpha_1 y'_{t-1} + \alpha_2 y'_{t-2} + \dots + \alpha_p y'_{t-p} & \left. \vphantom{y'_t} \right\} \text{AR}(p) \\ + \\ \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q} & \left. \vphantom{\epsilon_t} \right\} \text{MA}(q) \end{aligned}$$

Where $y'_t = y_t - y_{t-1}$

How do we
choose the
orders p
and q ?



Box—Jenkins method

Use ACF and **P**ACF to
determine p and q



How do we find temporal dependencies?

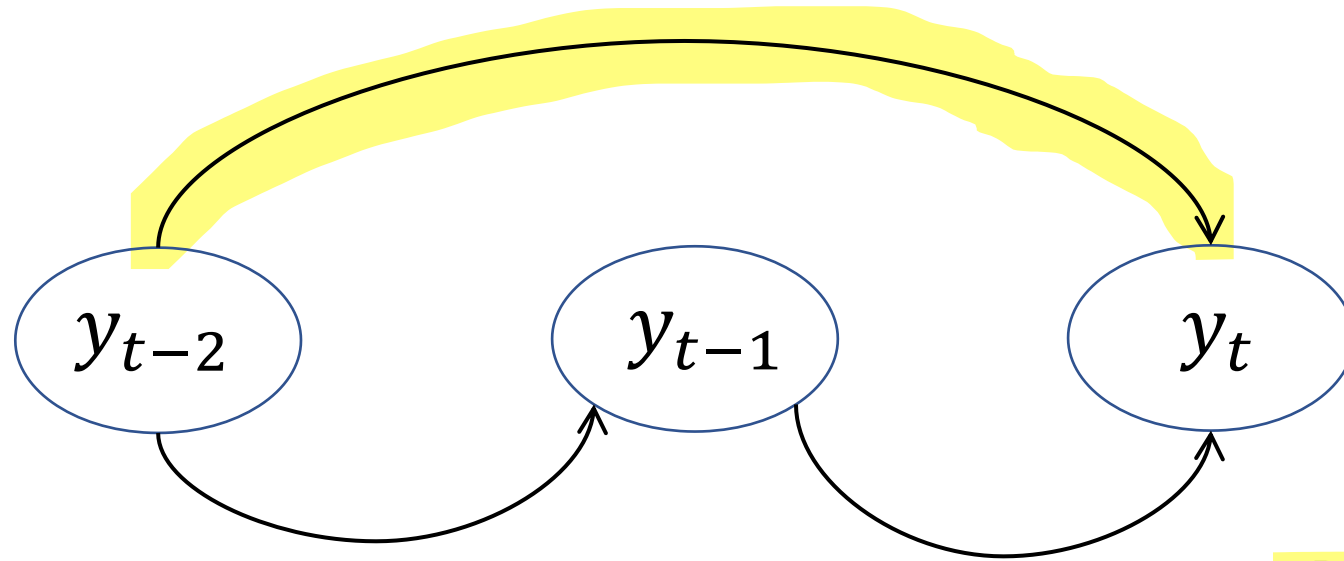
There are few concept to recall/understand:

- Correlation ✓
- Autocorrelation ✓
- Autocorrelation Function ✓
- **Partial Autocorrelation Function**



Recall

Partial Autocorrelation



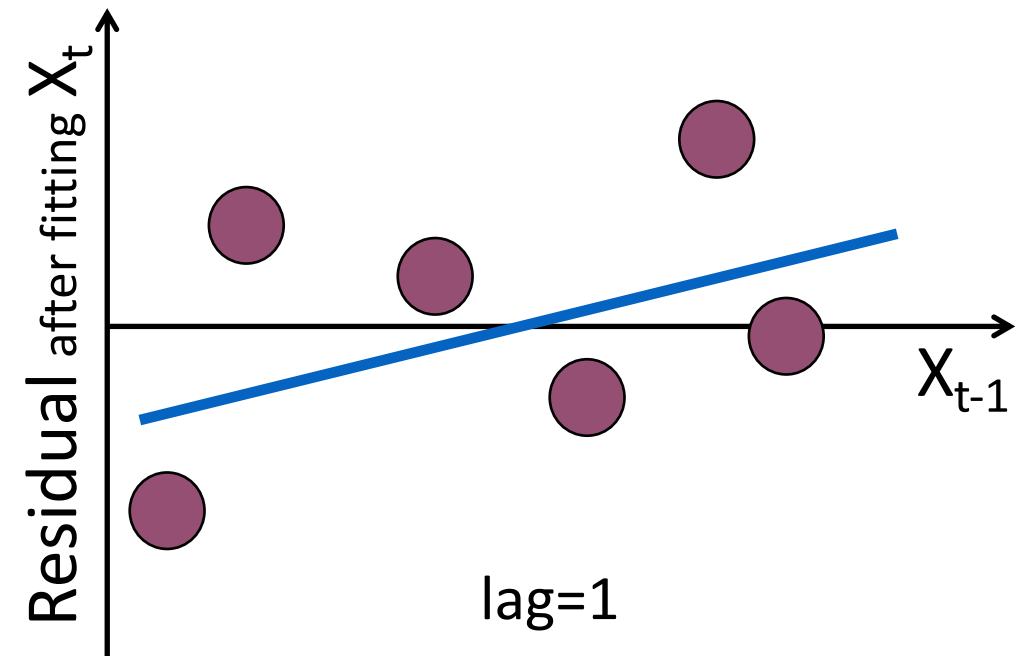
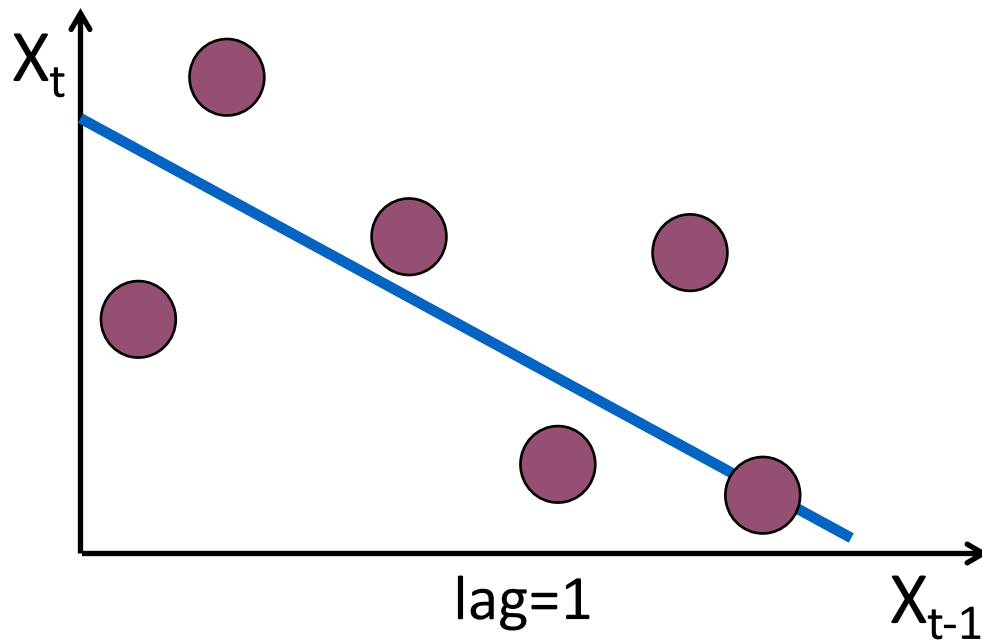
Direct correlation

$$ACF(y_{t-2}, y_t) \begin{cases} y_{t-2} \rightarrow y_t \\ y_{t-2} \rightarrow y_{t-1} \rightarrow y_t \end{cases}$$

Indirect correlation

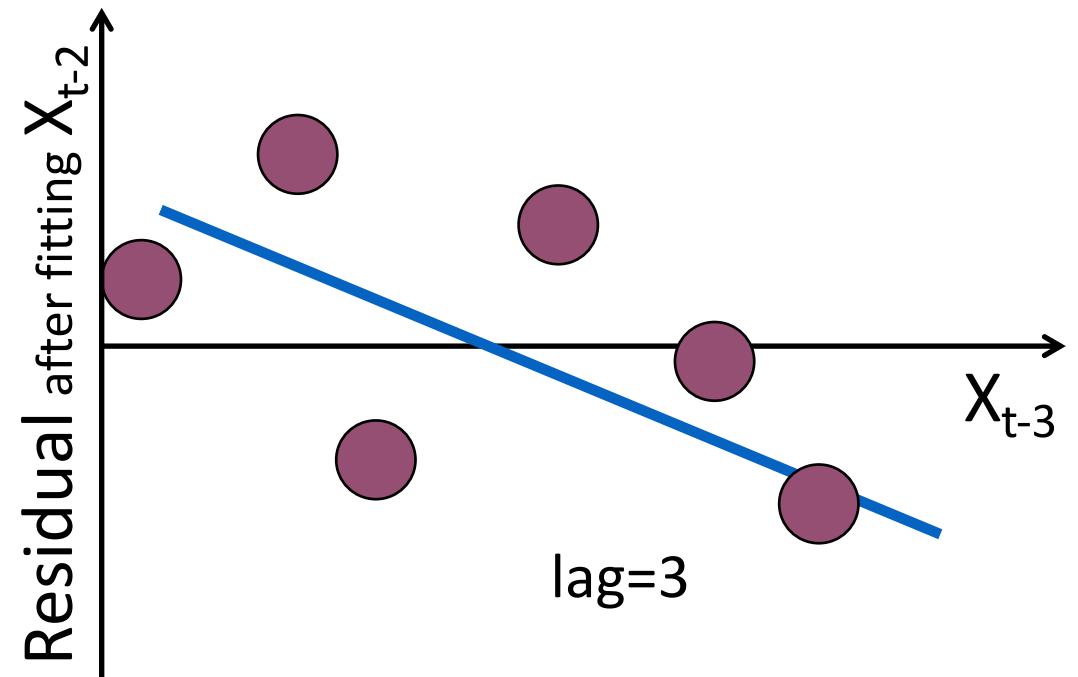
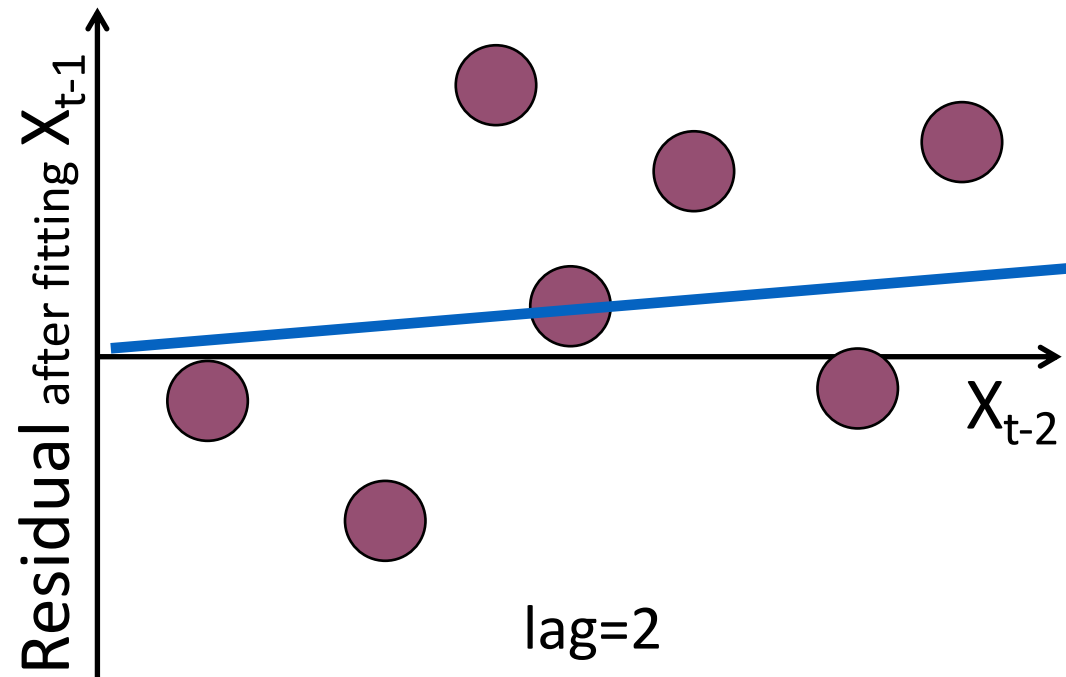
Partial Autocorrelation at lag 1

- **Partial Autocorrelation at lag 1** describes the degree of **similarity between the residual after fitting** of a given time series **and a version of itself lagged once**.

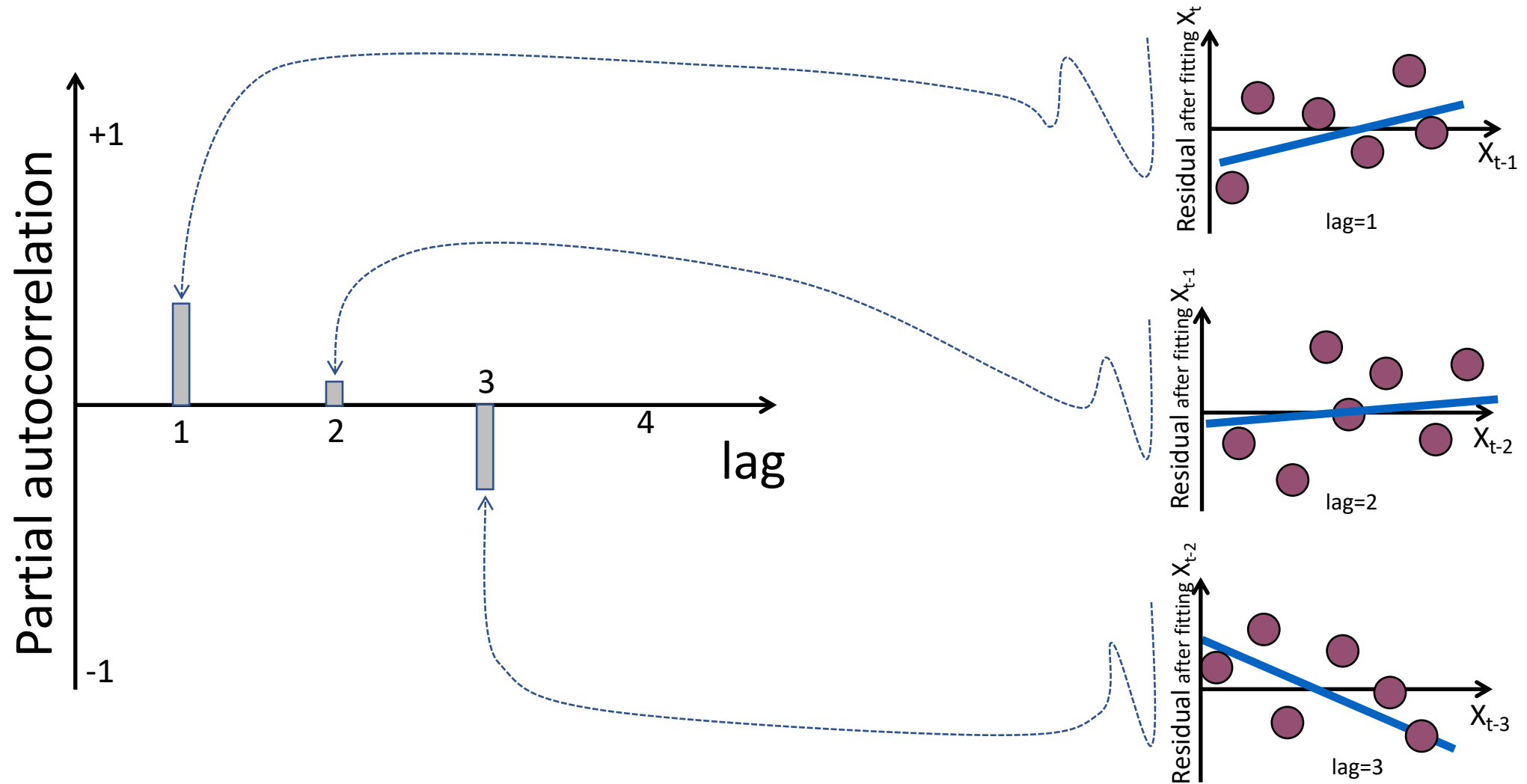


Partial Autocorrelation at lag k

- **Partial autocorrelation** describes the degree of similarity between the residual of fitting given time series and a lagged version of itself over successive time intervals.



Partial Autocorrelation Function (PACF)



When PACF is equal to ACF

The Partial Autocorrelation Function (PACF) **at lag 1** does not consider any intermediate observations; it is essentially the correlation between observations y_t and y_{t-1} while removing the effect of all intermediate observations.

$$PACF(1) = ACF(1) = \text{corr}(y_t, y_{t-1})$$

In this case, $PACF(1)$ is indeed equivalent to $ACF(1)$, as both represent the correlation between two consecutive observations without the influence of other intermediate observations.

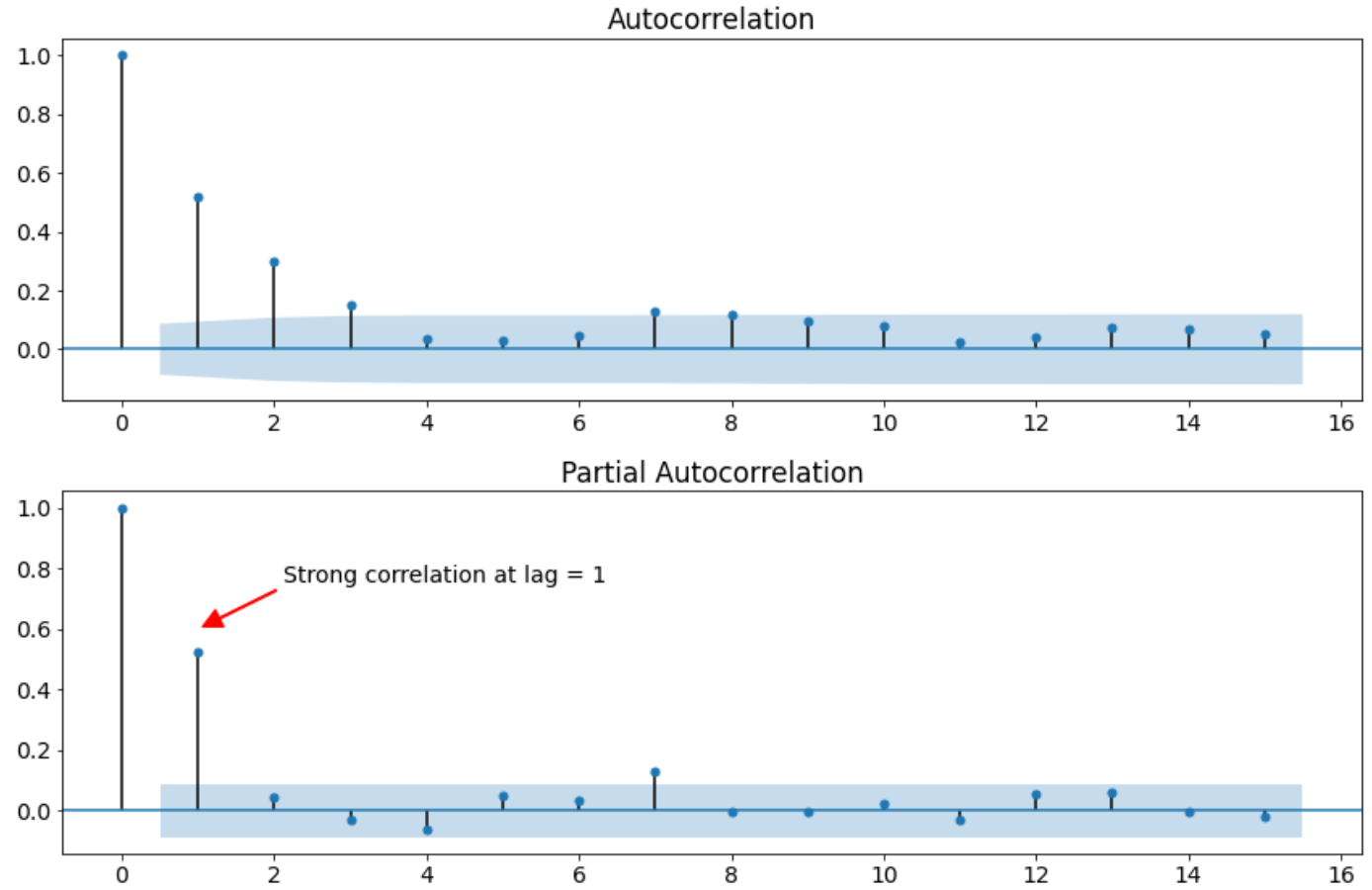
Use ACF and PACF to determine p and q

- The model is AR if the ACF tails off after a lag and has a hard cut-off in the PACF after a lag. This lag is taken as the value for p .
- The model is MA if the PACF tails off after a lag and has a hard cut-off in the ACF after the lag. This lag value is taken as the value for q .
- The model is a mix of AR and MA if both the ACF and PACF tail off.

Model	ACF	PACF
ARMA($p,0$)	Tails off after some lag	Cut-off after p lags
ARMA(0, q)	Cut-off after q lags	Tails off after some lag
ARMA(p,q)	Tails off	Tails off

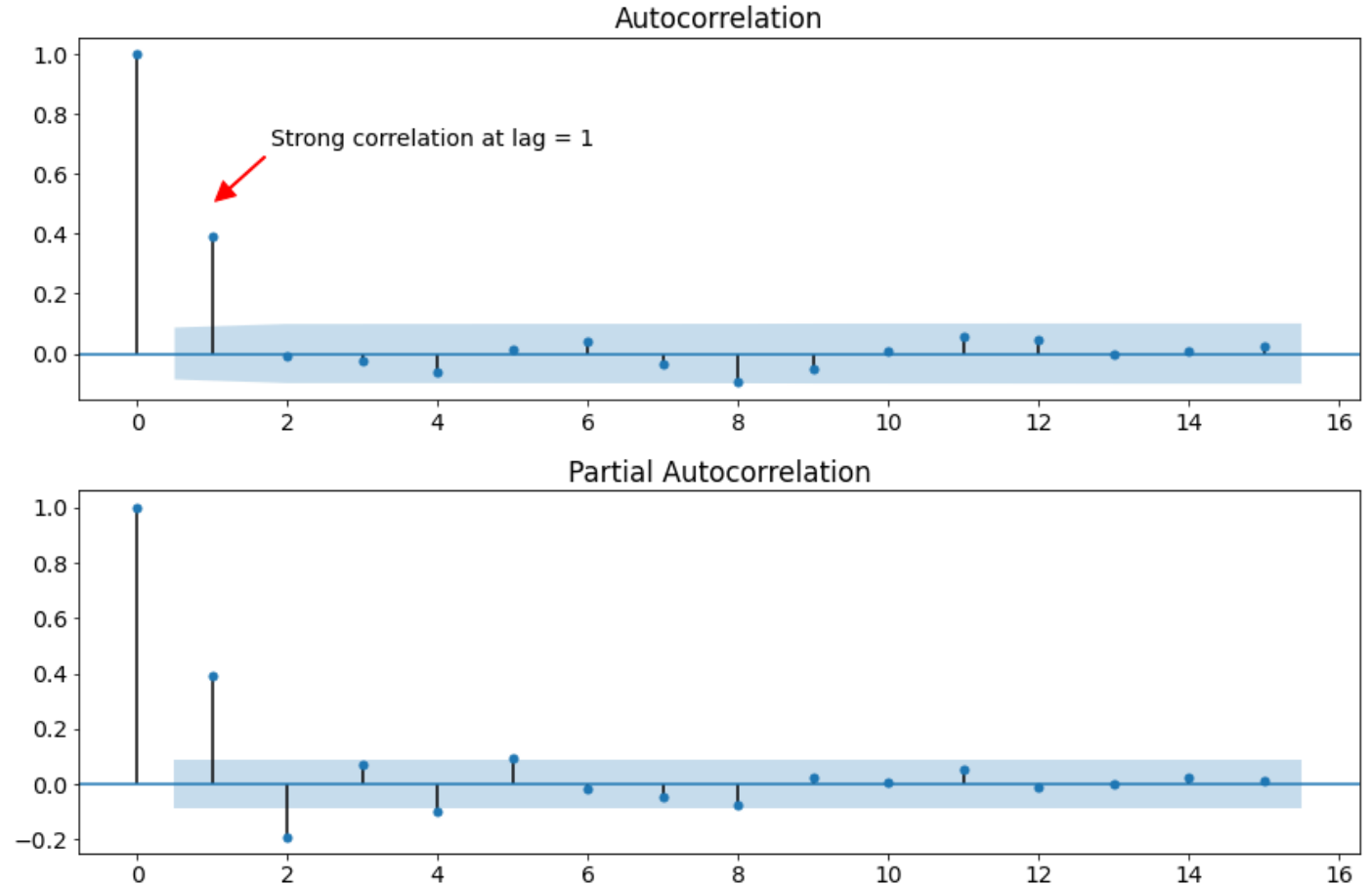
P and q order estimation: 1st example

- Autocorrelation: tails off
- Partial Autocorrelation: cut-off after 1 lag
- The model is :
 $AR(1) = ARMA(1,0)$



P and q order estimation: 2nd example

- Autocorrelation:
cut-off after 1 lag
- Partial Autocorrelation:
tails off
- The model is :
 $MA(1) = ARMA(0,1)$

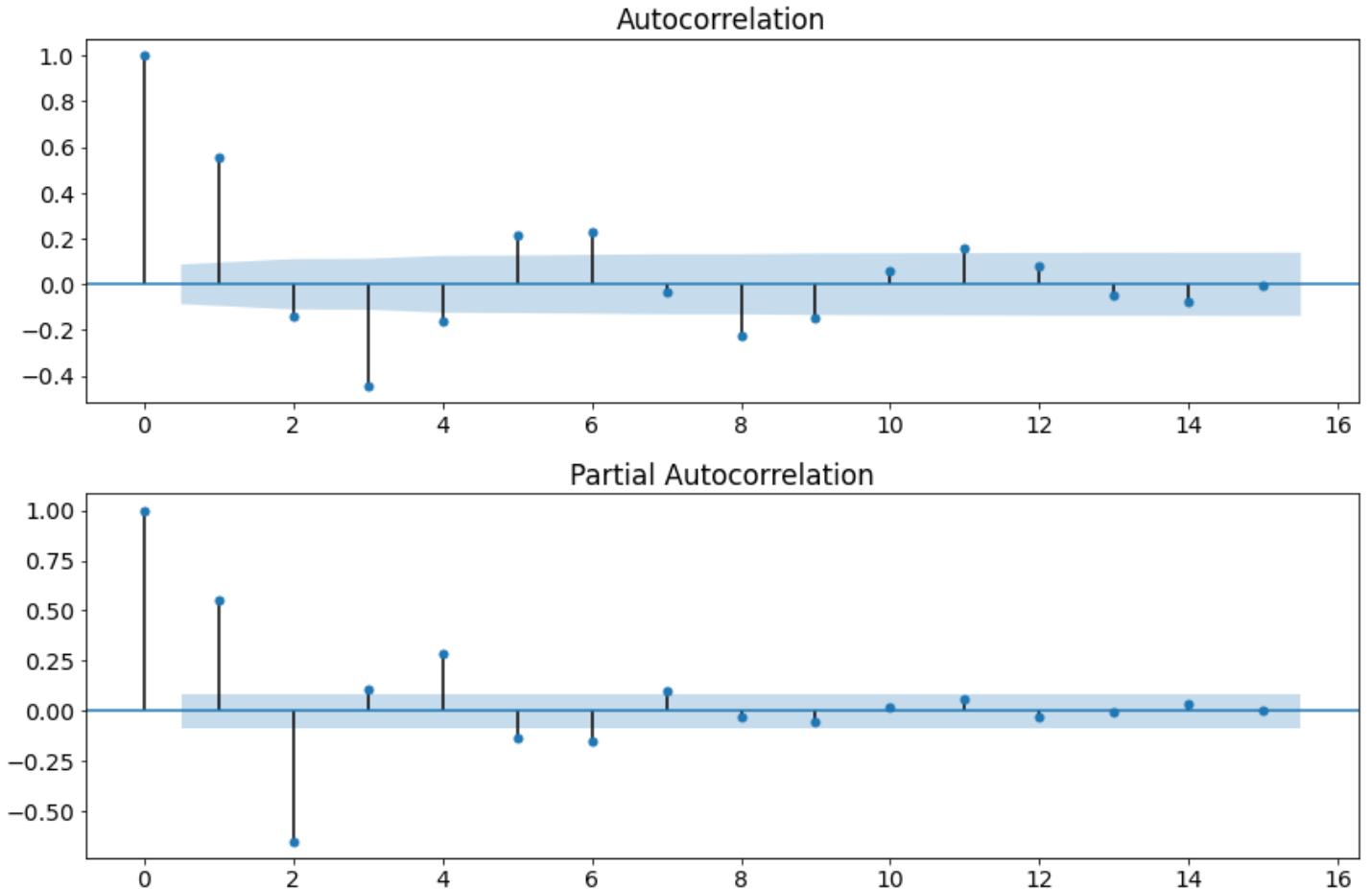


P and q order estimation: 3rd example

- Autocorrelation: tails off
- Partial Autocorrelation: tails off
- The model is :



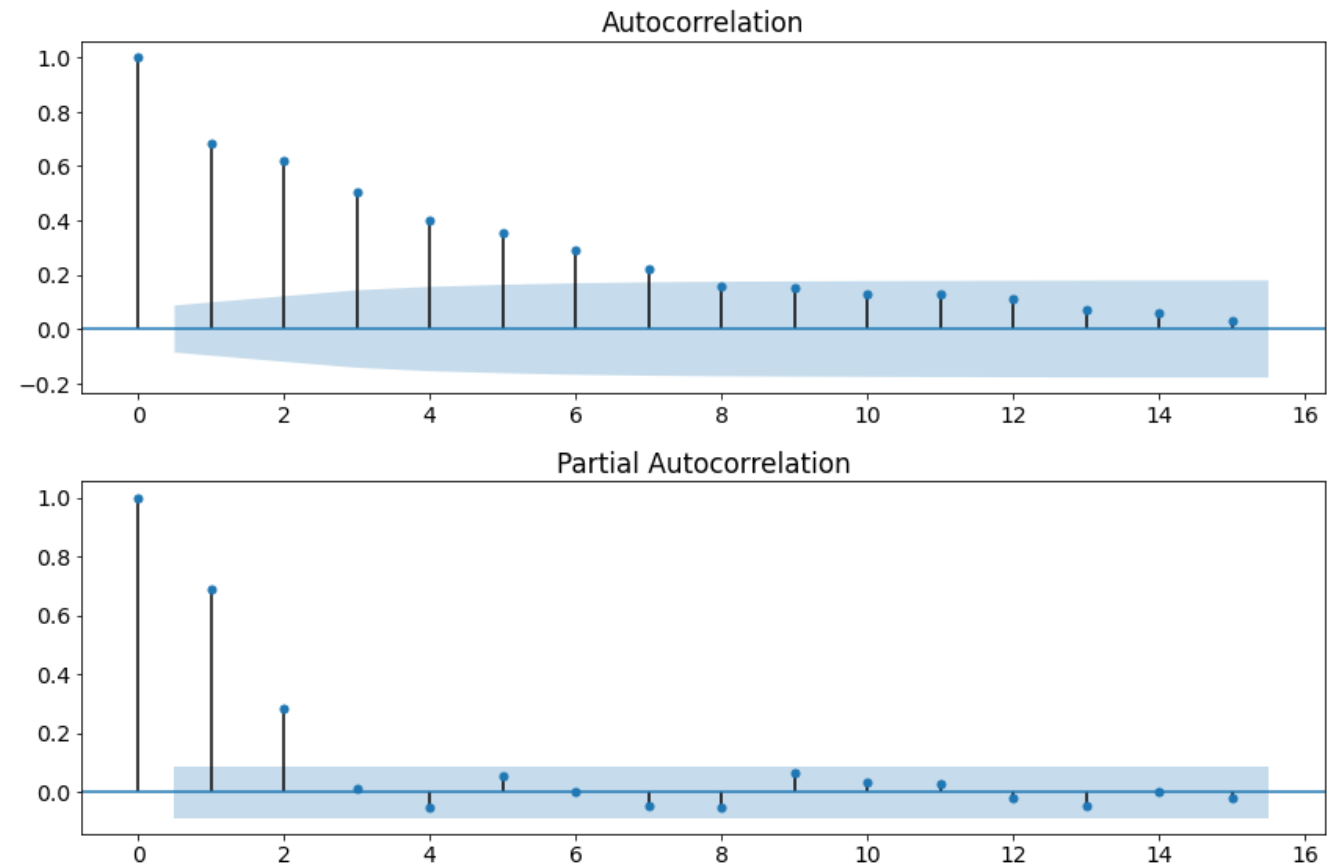
*the original process is an ARMA (2,2)



Order estimation test

Which model best characterises the process?

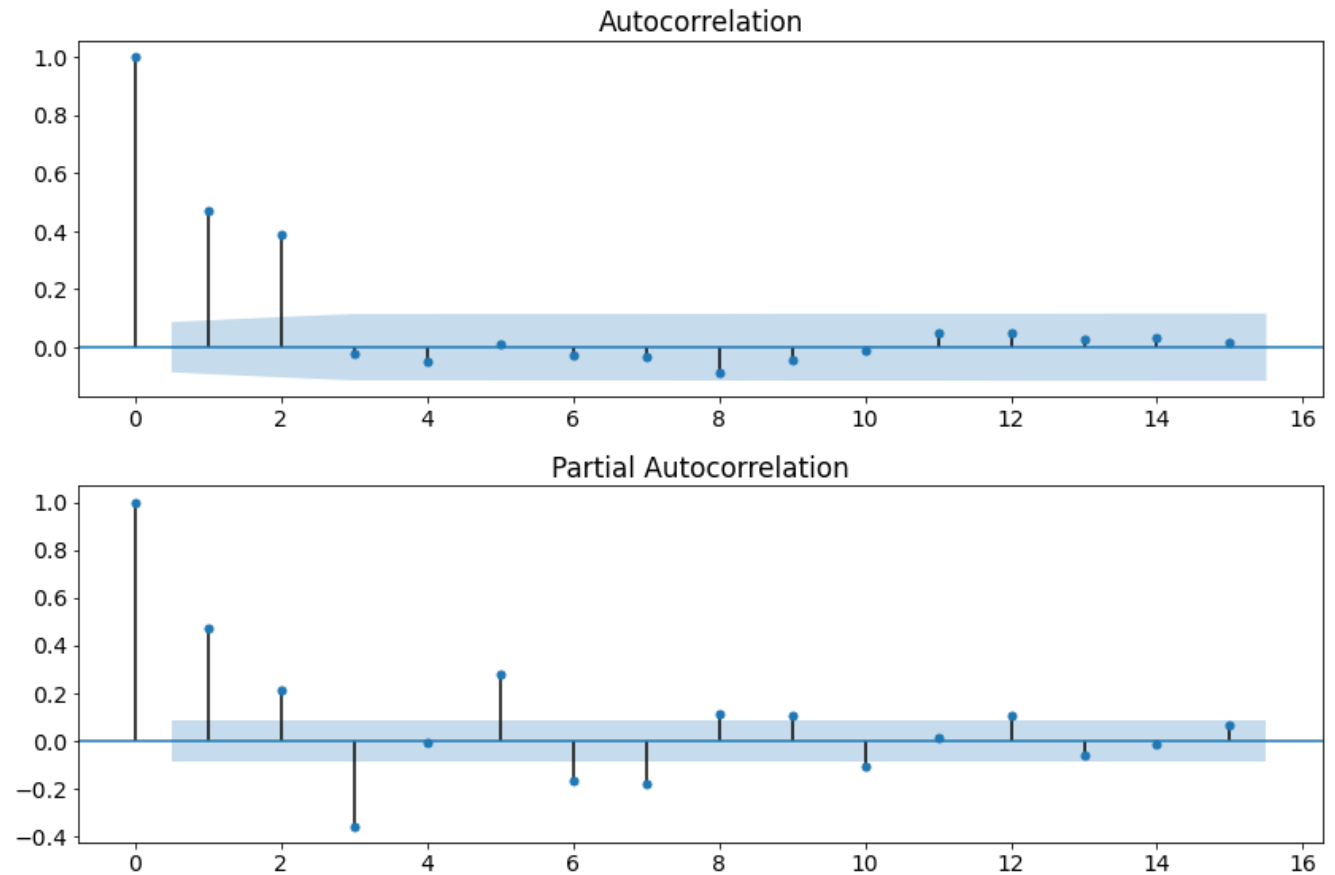
- ☐ an AR(1)
- ☐ an ARMA(2,1)
- ☐ an MA(2)
- ☐ an AR(2)



Order estimation test

Which model best characterises the process?

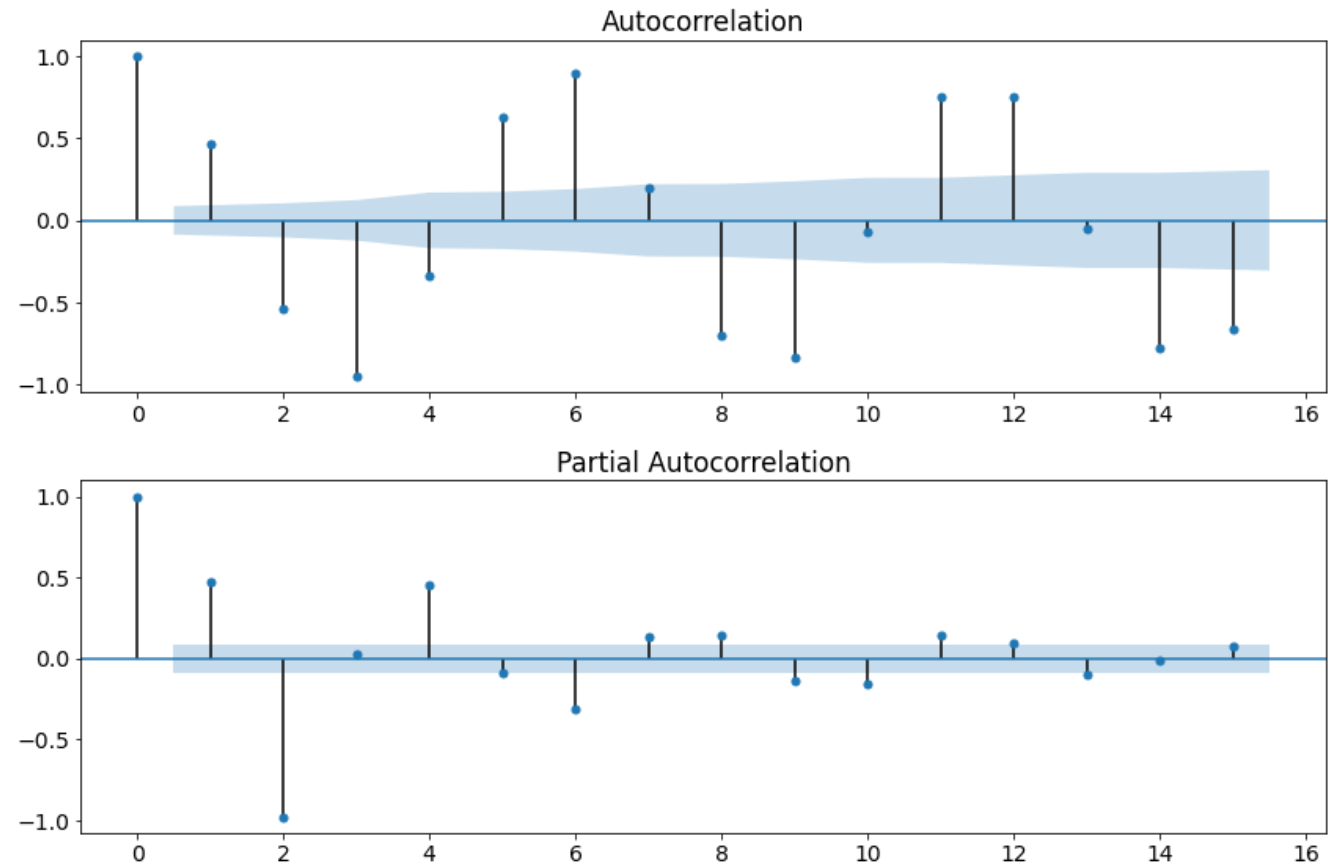
- ☐ an AR(1)
- ☐ an ARMA(2,1)
- ☐ an MA(2)
- ☐ an AR(2)



Order estimation test

Which model best characterises the process?

- ☐ an ARMA(1,1)
- ☐ an ARMA(2,2)
- ☐ an ARMA(3,2)
- ☐ an ARMA(3,3)

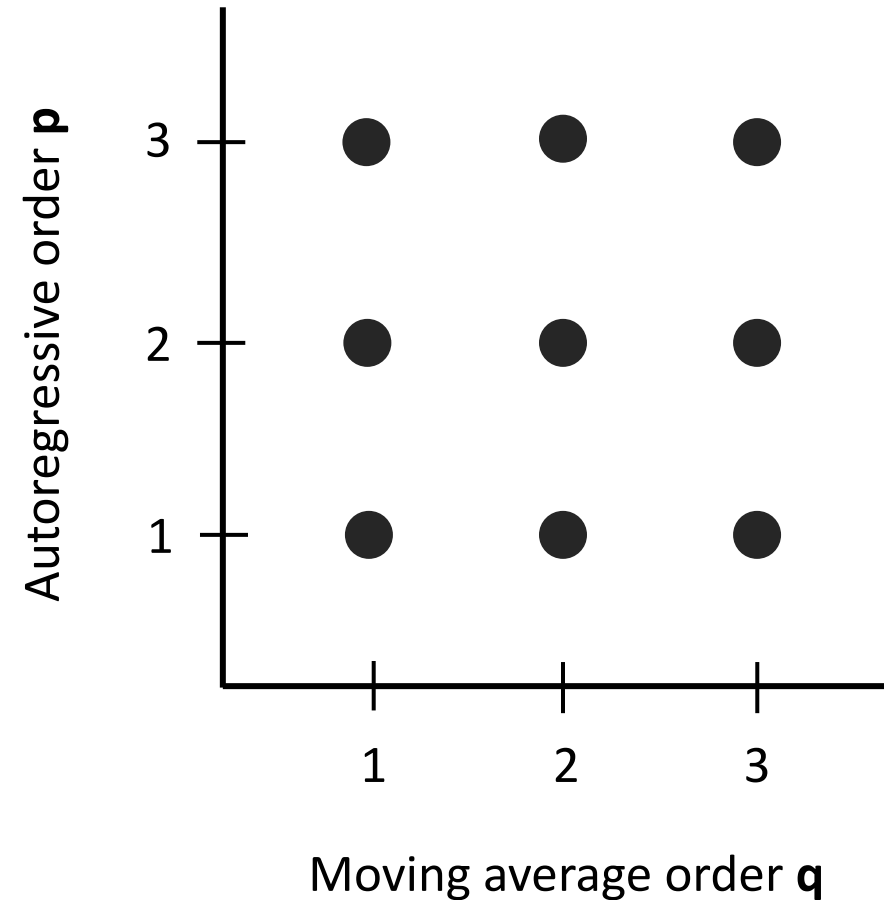


Beyond ACF and PACF: Grid search for choosing p and q order

Grid search

Autoregressive order $p = [1, 2, 3, \dots]$

Moving average order $q = [1, 2, 3, \dots]$



Which indicator to use? AIC and BIC!

Akaike Information Criteria (AIC) measures the quality of the model in terms of its goodness-of-fit to the data, its simplicity, and how much it relies on the tuning parameters. The formula for AIC is:

$$AIC = 2k - 2l$$

where l is a log-likelihood, and k is a number of parameters.

In addition to AIC, the **Bayesian Information Criteria** (BIC) uses one more indicator n that defines the number of samples used for fitting. The formula for BIC is:

$$BIC = k \log n - 2l$$

All information available in ARIMA results

SARIMAX Results

```
=====
Dep. Variable:          y          No. Observations:          1000
Model:                ARIMA(0, 0, 2)  Log Likelihood          9113.640
Date:                Fri, 20 Aug 2021  AIC                2794.875
Time:                10:44:25         BIC                2804.003
Sample:                0             HQIC               2789.279
                                - 1000
Covariance Type:        opg
=====
```

Goodness of fit

```
=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
ma.L1         0.6064         0.027     22.451     0.000         0.590         0.702
ma.L2         0.6249         0.022     19.047     0.000         0.471         0.579
sigma2        0.9461         0.042     22.621     0.000         0.864         1.028
=====
```

Parameters

```
=====
Ljung-Box (L1) (Q):          0.27   Jarque-Bera (JB):          7.04
Prob(Q):                    0.60   Prob(JB):              0.05
Heteroskedasticity (H):      0.91   Skew:                  0.19
Prob(H) (two-sided):         0.41   Kurtosis:              3.08
=====
```

Residuals

How to read ARIMA results

General information

- Dep. Variable
- Model
- Sample
- Covariance Type

SARIMAX Results

Dep. Variable: y

Model: ARIMA(0, 0, 2)

Date: Fri, 20 Aug 2021

Time: 10:44:25

Sample: 0

- 1000

Covariance Type: opg

No. Observations: 1000

Log Likelihood 9113.640

AIC 2794.875

BIC 2804.003

HQIC 2789.279

	coef	std err	z	P> z	[0.025	0.975]
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Goodness of fit

Parameters

Residuals

How to read ARIMA results

Goodness of fit

- Log likelihood
- AIC
- BIC
- HQIC

SARIMAX Results

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Date: Fri, 20 Aug 2021

Time: 10:44:25

Sample: 0

- 1000

Covariance Type: opg

No. Observations: 1000

Log Likelihood 9113.640

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Prob(JB): 0.05

Skew: 0.19

Kurtosis: 3.08

Goodness of fit

Parameters

Residuals

How to read ARIMA results

Parameters

- ma.L1 and ma.L2
- coeff
- std err
- z and $p > |z|$
- [0.025 – 0.975]
- sigma2

SARIMAX Results

Dep. Variable:	y	No. Observations:	1000
Model:	ARIMA(0, 0, 2)	Log Likelihood	9113.640
Date:	Fri, 20 Aug 2021	AIC	2794.875
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Covariance Type:	opg		

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Prob(Q):	0.60	Prob(JB):	0.05
Heteroskedasticity (H):	0.91	Skew:	0.19
Prob(H) (two-sided):	0.41	Kurtosis:	3.08

Goodness of fit

Parameters

Residuals

How to read ARIMA results

Residuals

- Ljung-Box
- Jarque-Bera

SARIMAX Results

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Model: ARIMA(0, 0, 2)

Date: Fri, 20 Aug 2021

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- 1000

Covariance Type: opg

No. Observations: 1000

Log Likelihood 9113.640

AIC 2794.875

BIC 2804.003

HQIC 2789.279

	coef	std err	z	P> z	[0.025	0.975]
ma.L1	0.6064	0.027	22.451	0.000	0.590	0.702
ma.L2	0.6249	0.022	19.047	0.000	0.471	0.579
sigma2	0.9461	0.042	22.621	0.000	0.864	1.028

Ljung-Box (L1) (Q): 0.27

Prob(Q): 0.60

Heteroskedasticity (H): 0.91

Prob(H) (two-sided): 0.41

Jarque-Bera (JB): 7.04

Prob(JB): 0.05

Skew: 0.19

Kurtosis: 3.08

Goodness of fit

Parameters

Residuals

Best practice

If a model is parsimonious, ...



... it makes the best forecast!

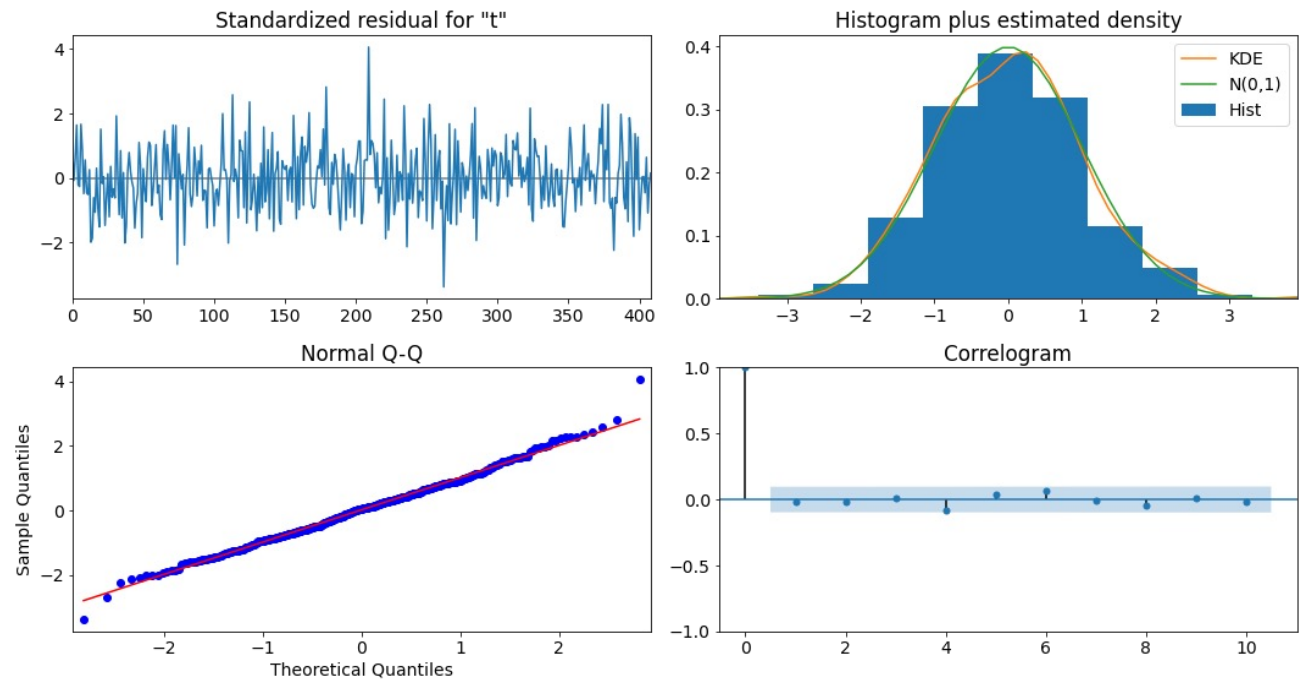


Parsimonious model and model diagnosis

- **Start** with a model that uses the **minimum p and q**
- **Diagnose** the residuals
 - **verifying** if they have a **normal distribution**
 - **checking-out** the **ACF** and **PACF** of the **residuals** to see if further temporal dependency are present
 - **small** autocorrelations and partial autocorrelations → model **OK**
 - **large** autocorrelations → **increase p and/or q**
- Criticism: this approach fails to identify if the model is too big

Model diagnostic

Test	Good fit
Standardized residual	There are no obvious patterns in the residuals
Histogram plus kde estimate	The KDE curve should be very similar to the normal distribution
Normal Q-Q	Most of the data points should lie on the straight line
Correlogram	95% of correlations for lag greater than zero should not be significant



ARMA(p,q) order estimation

1. Generate synthetic data
2. Plot ACF e PACF
3. Estimate p and q
4. Diagnose the model
5. Repeat until you find the “parsimonious” model



Lesson learned: Long-term forecast challenge

- Over time, errors accumulate, leading to straight-line predictions for extended horizons.
- Beyond the limits of order p and q , predictions tend to converge into a constant value due to reliance on forecasted values.
- **Prolonged predictions may result in a loss of accuracy**, often manifesting as a linear trend or stable value over time.

Seasonal ARIMA

$SARIMA(p, d, q) \times (P, D, Q)S$

- p = non seasonal AR order
- d = non seasonal differencing
- q = non seasonal MA order
- P = seasonal AR order
- D = seasonal differencing
- Q = seasonal MA order
- S = seasonality

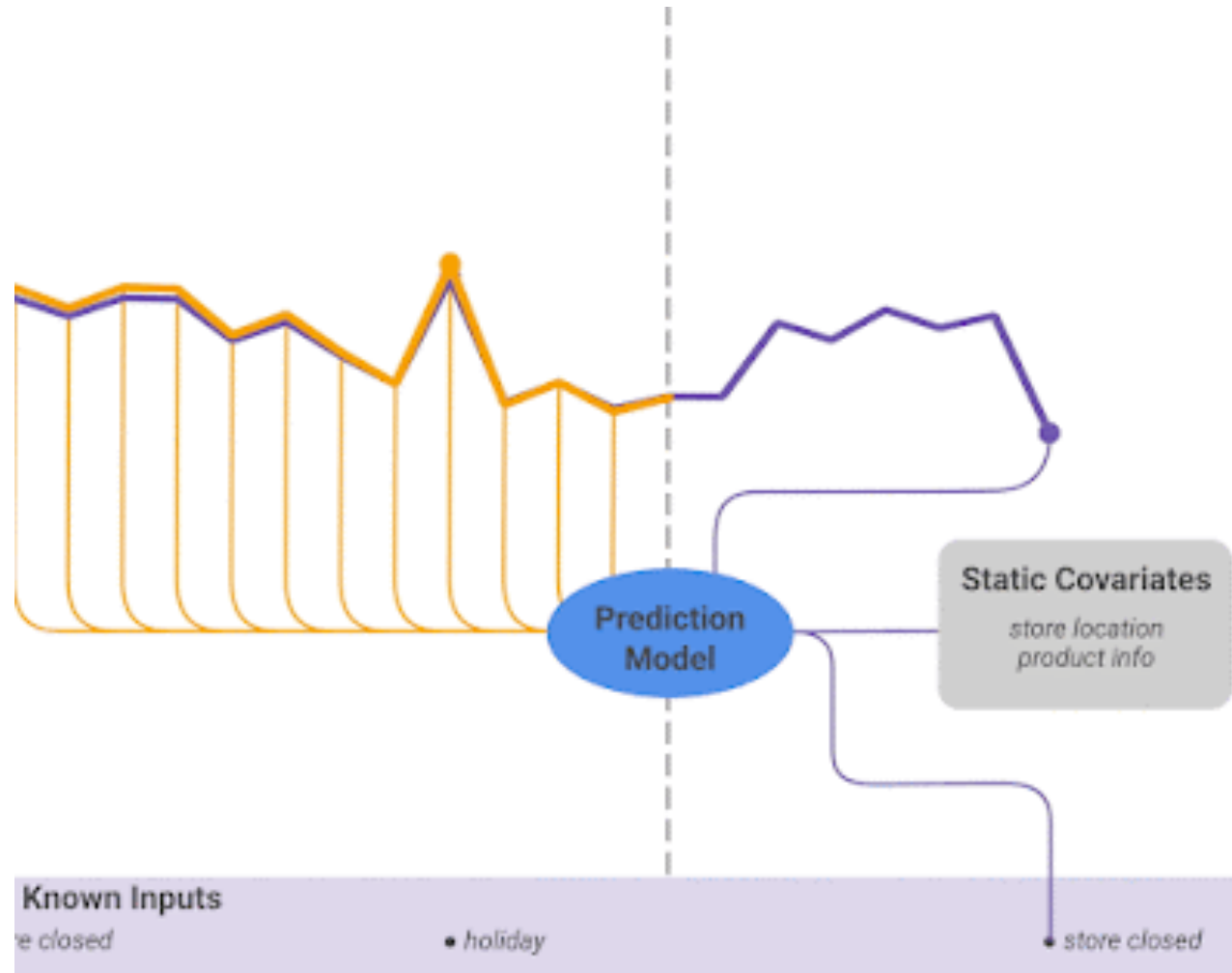


SARIMA Formula

This model is very similar to the ARIMA model, except that there is an **additional set of autoregressive and moving average components** for the seasonality component (of period s).

$$y_t = c + \sum_{n=1}^p \alpha_n y_{t-n} + \sum_{n=1}^q \theta_n \epsilon_{t-n} + \sum_{n=1}^P \phi_n y_{t-sn} + \sum_{n=1}^Q \eta_n \epsilon_{t-sn} + \epsilon_t$$

What about using external data (covariates)? SARIMAX!



SARIMAX

SARIMAX includes all the components of SARIMA but with an extension to **incorporate r exogenous variables**.

Exogenous variables are external factors that can impact the time series but are not influenced by it.

For example, in the context of sales forecasting, the number of advertising dollars spent could be an exogenous variable.

$$y_t = c + \sum_{n=1}^p \alpha_n y_{t-n} + \sum_{n=1}^q \theta_n \epsilon_{t-n} + \sum_{n=1}^r \beta_n x_{n_t} + \sum_{n=1}^P \phi_n y_{t-sn} + \sum_{n=1}^Q \eta_n \epsilon_{t-sn} + \epsilon_t$$

SARIMA and SARIMAX forecasting methods

1. Load the air passengers' time series
2. Fit several models including SARIMA
3. Find out the best model and discuss why
4. Compare SARIMA and SARIMAX



Time-series Temporal Dependence & ARIMA

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